DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

Proposal for MTech (CSE) Curriculum

A. M.Tech (CSE)

- Students for M.Tech Degree would be required to complete minimum of 48 PG credits and 6 credits for Bridge courses (BC) to earn the Degree.
    - The requirement of Bridge Credits may be fully or partially waived off based on recommendation of the DRC keeping in view prior background of the student.
- The department proposes to offer two types of MTech Programme
    - MTech with coursework option with 10 credit project component (Minor project+Major Project Part I) and
    - MTech with dissertation (a research oriented) with 24 credit project work over 3 semesters.
        - To opt for dissertation based M.Tech, student must have a minimum CGPA of 7.5 after 2nd semester and MTP-I with B and above grade.
        - In exceptional cases DRC may waive the CGPA requirement.
        - Students may choose specialization based on credits earned in a particular specialization (refer section C)

B. Structure: M.Tech Options I & II

I. Bridge Credits: 0/6 As per recommendation of DRC

II. Program Core: Min 24 Cr

COL701: Software Systems Lab	0-0-6	3 cr
COL702: Advanced Data Structures	3-0-2	4 cr
COL765: Logic & Functional Programming	3-0-2	4 cr
One of the following three		
COL703: Logic for CS (LCS)	3-0-2	4 cr /
/ COL 726: Numerical Algorithms (NA)	3-0-2	4 cr /
/ COL705: Theory of Computation (TOC)	3-0-0	3 cr
Project Component		
COP 891: **Minor Project**	0-0-6	3cr
COD 892: M.Tech Project Part I (MTP-I)	0-0-14	7cr

III. Program Electives (PE):

a) **MTECH OPTION I: COURSEWORK OPTION:** (About 6 courses) **:** **Min 24 Cr**

- Program Elective (PE) courses from 700/800 level: 20 cr (Min)
- One PE / Open Category elective
  (from 700/800 level, outside the department): 4 cr (Max)

b) MTECH OPTION II: DISSERTATION BASED: 24 Cr

• COP893: MTP-II (DISSERTATION):	0-0-28	14 cr
• PE courses		10 cr

Requirements:

- Student must complete a thesis as a part of Major Project Part II (MTP-II) of 14 credits. MTP-II may be in continuation with MTP-I.
- In addition, 10 credits of PE courses (about 3 courses).

C. Departmental Streams/Specialization for Option II

Student may opt for one of the following Specializations in his/her degree.

- Architecture & Embedded Systems (AES)
- Graphics & Vision (GV)
- Software Systems (SS)
- Theoretical Computer Science (TH)
- Data Analytics & AI (DAAI)
- Applications & IT (ITA)

Requirements:

- At least 6 credits in PE from the specialization.
- MTP-I and MTP-II must be relevant to the specialization chosen.
- The list of available electives and relevant specialization (Refer Annexure A).

D. Indicative Course Schedule

Sem	Lecture Course I	Lecture Course II	Lecture Course III / Project	Lecture Course IV	Comments	L-T-P/Cr
I	COL 702: Advanced Data Structures 4Cr	BC1 4Cr / 3Cr	COL 765: Logic and Functional Programming 4 Cr	COL 701: Software Lab 0-0-6 3Cr		11 - 14
II	One of these LCS/NA/TOC 4 Cr/ 3Cr	PE1 4Cr / 3Cr	COL 891: Minor Project 3-0-0 3Cr	BC2 4Cr /3Cr		11 - 15
III	PE2 4Cr	PE3 4Cr	COL 892: MTech Project -1 0-0-14 7Cr			15
M.Tech Option I COURSEWORK						
IV	PE4 4Cr	PE5/OC 4Cr	PE6 4Cr			12
M.Tech Option II Thesis						
IV	MTP2 0-0-28					14
Summer		MTP2 (Contd)				
Complete MTech Requirement					Total Credits	≥ 48

Proposal for Dual Programme Curriculum

Department Linked EA/ES & BS Courses

1. Probability and Stochastic Process (Maths) 4cr
2. Signals & Systems (EE) – 4cr
3. Material Science (PHY) 3Cr
    - PHL 111: Principles of Electronic Materials OR
    - PHL 113: Physics of Nanomaterials
4. One of the following courses – 3-4cr
    - MAL 256 Modern Algebra
    - MAL 255 Linear Algebra: Theory, Methods and Applications
    - MAL 210 Optimization Methods and Applications

Total 14-15* cr

*If a student does 15 credits, then extra 1 credit will count towards DE.

	PROPOSED Departmental Core	**LTP**	**Cr**
1	COL 106: Data Structures & Algorithms	3-0-4	5
2	COL 202: Discrete Structures	3-1-0	4
3	COL 215: Digital Logic & Design	3-0-4	5
4	COL 216: Computer Architecture	3-0-2	4
5	COL 226: Programming Languages	3-0-4	5
6	COP 290: Design Practices	0-0-6	3
7	COL 331: Operating Systems	3-0-4	5
8	COL 333: Principles of Artificial Intelligence*	3-0-2	4*
9	COL 334: Computer Networks	3-0-2	4
10	COL 351: Algorithm Design & Analysis	3-1-0	4
11	COL 352: Intro to Automata & Theory of Computation	3-0-0	3
12	COL 362: Introduction to Database Management Systems*	3-0-2	4*
13	COL 380: Intro to Parallel & Distributed Programming	2-0-2	3
	Total 12 courses + 1 Design Lab		49

*Note: Student must do one of CSL333 or CSL362 as part of DC (if he does both, second course will count as DE)

Requirements for B.Tech Degree in CSE for Dual Programme:

- Department Core + 11 Credits of DE
    - 3 Courses of 3-4 credits each
- Overall Credit Requirements:
    - Institute core 55 + Dept Core 49 + 14 (PL-EAS) = 118core

– DE 11 + 10 (OC) = 21
– Total 139cr + 15 credits of Non Graded Core

Requirements for M.Tech CSE for Dual Degree:

- Students for Dual Degree would be required to complete total of 139 UG + 46 PG credits to earn the Dual Degree

  Total 185 cr + 15 Non-Graded
- Continuation to M.Tech in Dual Programme would be subject to maintenance of **Minimum GPA 6.75** in UG part of the programme.

Structure:

Institute Core 55 + DC 49 + PLES 14 + DE 11 + OC 10= 139

+

Program Core (PC): 32 Cr

COL 703: Logic for CS	3-0-2	4 cr
COL 726: Numerical Algorithms	3-0-2	4 cr
COP 891: Minor Project	0-0-6	3 cr
COP 892: MTech Project Part I	0-0-14	7 cr
COP 893: MTech Project Part II	0-0-28	14 cr

+

Program Electives (PE): 14 Cr

4 Electives from list of departmental 700/800 level electives

Specializations offered

- Advanced Topics in CS
- Architecture & Embedded Systems (AES)
- Graphics & Vision (GV)
- Software Systems (SS)
- Theoretical Computer Science (TH)
- Data Analytics & AI (DAAI)
- Applications & IT (ITA)

- The list of available electives and relevant specialization is given in Annexure A.

CS5: Indicative Course Schedule: (Credits completed in 1st year: 34 Credits of Institute Core)

Sem	Lecture Course I	Lecture Course II	Lecture Course III	Lecture Course IV	Lecture CourseV	other	L-T-P/Cr
III	COL 202: Discrete Structures 3-1-0 4 Cr/ DC	COL 215: Digital Logic & Design 3-0-4 5 Cr /DC	COL106: Data Structures 3-0-4 5 Cr /DC	Material Science [Physics] 3 [PLES2]	Probability & Stochastic Processes 3-1-0 [PLES1]	CON100 0-0-2 Non-Graded	21
IV	COL 226: Prog. Languages 3-0-4 5 Cr /DC	COL216: Computer Architecture 3-0-2 4 Cr/ DC	EEL205: Signals and Systems 3-1-0 [PLES3] 4Cr	Env (2-0-0) 2 Cr	HU1 4 Cr	COP 290: Design Practices in CS 0-0-6 3 Cr /DC	22
V	COL 333: Principles of AI 3-0-2 ~~[DC]~~ DE1	COL 331: Operating Systems 3-0-4 5 Cr /DC	COL 352: Theory of Computing 3-0-0 3 Cr / DC	Biology (3-0-2) 4 Cr	HU2 4 Cr		20
VI	COL 362: Data Base Mgmt Systems 3-0-2 ~~[DC]~~ DE1	COL 334: Computer Networks 4 Cr / DC	COL 351: Analysis & Design of Algorithms 3-1-0 4 Cr /DC	Math 3-0-0 [PLES 4]	HU3 4 Cr	COL 380: Intro to Parallel & Dist. Programming 2-0-2 3 Cr / DC	22
VII	DE2 3 Cr	DE3 4 Cr	COL 703: Logic for CS 4 Cr / PC		OC1 3 Cr	COP 891: Minor Project 3 Cr/ PC	17
VIII	PE1 3 Cr	COL 726: Numerical Algorithms 4 Cr / PC	OC2 3 Cr	PE2 3 Cr	HU4 3 Cr	PE3 3 Cr	19
IX	PE4 3 Cr	PE5 3 Cr	OC3 3 Cr			MTP Part-I 0-0-14 7 Cr	16
X						MTP Part-II 0-0-28 14 Cr	14

Total 185

Annexure A

List of PG courses

Course Number & Title	Status	L-T-P	Credits	Stream
COL632: Introduction to Data Base Systems	BC	3-0-2	4	
COL633: Resource Management in Computer Systems	BC	3-0-2	4	
COL671: Artificial Intelligence	BC	3-0-2	4	
COL672: Computer Networks	BC	3-0-2	4	
COP701: Software Systems Laboratory		0-0-6	3	
COL702: Advanced Data Structure	UG/PG	3-0-2	4	TH/Core
COL703: Logic for CS	UG/PG	3-0-2	4	TH/Core
COL705: Theory of Computation	UG/PG	3-0-0	3	TH
COL718: Architecture of High Performance Computers	UG/PG	3-0-2	4	AES
COL719: Synthesis of Digital Systems	UG/PG	3-0-2	4	AES
COL722: Introduction to Compressed Sensing	UG/PG	3-0-0	3	ITA
COL728: Compilers	UG/PG	3-0-3	4.5	SS
COL729: Compiler Optimization	UG/PG	3-0-3	4.5	SS
COL724: Advanced Computer Networks	UG/PG	3-0-2	4	SS
COL726: Numerical Algorithms	UG/PG	3-0-2	4	TH
COL730: Parallel Programming	UG/PG	3-0-2	4	SS&TH
COL732: Virtualization and Cloud Computing	UG/PG	3-0-2	4	SS
COL733: Cloud Computing Technology Fundamentals	UG/PG	3-0-2	4	SS
COL740: Software Engineering	UG/PG	3-0-2	4	SS
COD745:Minor Project	PG	0-0-6	4	SS
COL750: Foundations of Automatic Verification	UG/PG	3-0-2	4	TH
COL751: Algorithmic Graph Theory	UG/PG	3-0-0	3	TH
COL752: Geometric Computing	UG/PG	3-0-0	3	TH
COL753: Complexity Theory	UG/PG	3-0-0	3	TH
COL754: Approximation Algorithms	UG/PG	3-0-0	3	TH
COL756: Mathematical Programming	UG/PG	3-0-0	3	TH
COL757: Model Centric Algorithm Design	UG/PG	3-0-2	4	TH & ITA
COL758: Advanced Algorithms	UG/PG	3-0-2	4	TH
COL759: Cryptography & Computer Security	UG/PG	3-0-0	3	TH
COL760: Advanced Data Management	UG/PG	3-0-2	4	DAAI & ITA

COL762: Database Implementation	UG/PG	3-0-2	4	DAAI & ITA
COL765: Introduction o Logic and Functional Programming	UG/PG	3-0-2	4	DAAI & ITA
COL768: Wireless Networks	UG/PG	3-0-2	4	SS
COL770: Advanced AI	UG/PG	3-0-2	4	DAAI & ITA
COL772: Natural Language Processing	UG/PG	3-0-2	4	DAAI
COL774: Machine Learning	UG/PG	3-0-2	4	DAAI
COL776: Learning Probabilistic Graphical Models	UG/PG	3-0-2	4	DAAI
COL780: Computer Vision	UG/PG	3-0-2	4	GV
COL781: Computer Graphics	UG/PG	3-0-3	4.5	GV
COL783: Digital Image Analysis	UG/PG	3-0-3	4.5	GV
COL786: Advanced Functional Brain Imaging	UG/PG	3-0-2	4	DAAI, ITA
COL788: Embedded Computing	UG/PG	3-0-0	3	AES
COS799: Independent Study	PG	0-3-0	3	**
COL812: System Level Design and Modelling	UG/PG	3-0-0	3	AES
COL818 Principles of Multiprocessor Systems	UG/PG	3-0-2	4	AES
COL819: Advanced Distributed Systems	UG/PG	3-0-2	4	SS
COL821: Reconfigurable Computing	UG/PG	3-0-0	3	AES
COL829: Advanced Computer Graphics	UG/PG	3-0-2	4	GV
COL830: Distributed Computing	UG/PG	3-0-0	3	TH
COL831: Semantics of Programming Languages	UG/PG	3-0-0	3	TH
COL832: Proofs and Types	UG/PG	3-0-0	3	TH
COL851: Special Topics in Operating Systems	UG/PG	3-0-0	3	SS
COL852: Special Topics in Compilers	UG/PG	3-0-0	3	SS
COL860: Special Topics in Parallel Computation	UG/PG	3-0-0	3	SS & TH
COL861: Special Topics in Hardware Systems	UG/PG	3-0-0	3	AES
COL862: Special Topics in Software Systems	UG/PG	3-0-0	3	SS
COL863: Special Topics in Theoretical Computer Science	UG/PG	3-0-0	3	TH
COL864: Special Topics in Artificial Intelligence	UG/PG	3-0-0	3	DAAI
COL865: Special Topics in Computer Applications	UG/PG	3-0-0	3	ITA
COL866: Special Topics in Algorithms	UG/PG	3-0-0	3	TH
COL867: Special Topics in High Speed Networks	UG/PG	3-0-0	3	SS
COL868: Special Topics in Database Systems	UG/PG	3-0-0	3	DAAI

COL869: Special Topics in Concurrency	UG/PG	3-0-0	3	DAAI & ITA
COL870: Special Topics in Machine Learning	UG/PG	3-0-0	3	DAAI
COL871: Special Topics in programming languages & Compilers	UG/PG	3-0-0	3	SS
COL872: Special Topics in Cryptography	UG/PG	3-0-0	3	TH
COV877:Special Module on Visual Computing	UG/PG	1-0-0	1	GV
COV878: Special Module in Machine Learning	UG/PG	1-0-0	1	DAAI
COV879: Special Module in Financial Algorithms	UG/PG	2-0-0	2	TH
COV880: Special Module in Parallel Computation	UG/PG	1-0-0	1	SS
COV881: Special Module in Hardware Systems	UG/PG	1-0-0	1	AES
COV882: Special Module in Software Systems	UG/PG	1-0-0	1	SS
COV883: Special Module in Theoretical Computer Science	UG/PG	1-0-0	1	TH
COV884: Special Module in Artificial Intelligence	UG/PG	1-0-0	1	DAAI
COV885: Special Module in Computer Applications	UG/PG	1-0-0	1	ITA
COV886: Special Module in Algorithms	UG/PG	1-0-0	1	TH
COV887: Special Module in High Speed Networks	UG/PG	1-0-0	1	SS
COV888: Special Module in Database Systems	UG/PG	1-0-0	1	DAAI & ITA
COV889: Special Module in Concurrency	UG/PG	1-0-0	1	DAAI & ITA
COP891: Minor Project	PG	0-0-6	3	**
COP 892: Major Project Part 1	PG	0-0-14	7	**
COP 893: Major Project Part 2	PG	0-0-28	14	**
SIL765: Networks & System Security	UG/PG	3-0-2	4	SS
SIL769: Internet Traffic - Measurement, Modeling & Analysis	UG/PG	3-0-2	4	SS & ITA
SIL801: Special Topics in Multimedia System	UG/PG	3-0-0	3	GV & ITA
SIL802: Special Topics in Web Based Computing	UG/PG	3-0-0	3	ITA
SIV813: Applications of Computer in Medicines	UG/PG	1-0-0	1	ITA
SIV861: Information and Comm Technologies for Development	UG/PG	1-0-0	1	ITA
SIV864: Special Module on Media Processing & Communication	UG/PG	1-0-0	1	ITA
SIV895: Special Module on Intelligent Information Processing	UG/PG	1-0-0	1	ITA

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Introduction to Database Systems
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL632
6.	**Status** (*category* for *program*)	Bridge Course for MTech (MCS),

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	100% with COL362 Database Management Systems (Maths Dept)
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	Students done equivalent course are not allowed

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	S.K. Gupta, Maya Ramanath
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course will introduce the basics of Database Management Systems. While this area is now extremely broad, the bulk of the focus will be on traditional and foundational data management techniques. Building on these basics, the course will then introduce students to recent topics, which they can pursue in advanced courses.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Data models (ER, relational models, constraints, normalization), declarative querying (relational algebra, datalog, SQL), query processing/optimization (basics of indexes, logical/physical query plans, views) and transaction management (introduction to concurrency control and recovery). Overview of XML data management, text management, distributed data management. Course project to build a web-based database application.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Data models	6
2	Declarative querying with relational algebra and SQL	3
3	Practical databases (constraints, views, triggers)	4
4	Index structures	3
5	Introduction to query processing/optimization	7
6	Declarative querying with datalog and recursive SQL	2
7	Transaction management	6
8	Introduction to XML data management	3
9	Introduction to Text data management	4
10	Introduction to Distributed data management	4
11		
COURSE TOTAL ***(14 times 'L')***		42

16.	**Brief description of tutorial activities**

N.A.

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Data modeling	4
2	Learning to use Postgres	5
3	Learning to write SQL queries	9
4	Learning to read query plans	10
COURSE TOTAL ***(14 times 'P')***		28

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, "Database Systems: The Complete Book", 2nd edition, Prentice Hall, 2008.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Java, Perl/PHP,PostgreSQL
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Resource Management in Computer Systems
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL633
6.	**Status** (*category* for *program*)	Bridge Course for MTech (MCS),

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	COL331
8.2	Overlap with any UG/PG course of other Dept./Centre	Courses on OS offered by EE and Maths

9.	**Not allowed for** (indicate program names)	Students done equivalent course are not allowed

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course**	Subhashis Banerjee, Huzur Saran, Smruti Sarangi, Sorav Bansal, Kolin Paul
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To introduce the design and implementation of OS interfaces and understand the performance and modularity tradeoffs involved, provide experience with non-trivial event-driven and concurrent programming, study synchronization primitives and their uses, understand handling of disk, network and other devices.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Primary UNIX abstractions: threads, address spaces, filesystem, devices, interprocess communication; Introduction to hardware support for OS (e.g., discuss x86 architecture); Processes and Memory; Address Translation; Interrupts and Exceptions; Context Switching; Scheduling; Multiprocessors and Locking; Condition Variables, Semaphores, Barriers, Message Passing, etc.; Filesystem semantics, design and implementation; Filesystem Durability and Crash recovery; Security and Access Control

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to UNIX Abstractions	4
2	Hardware Support for OS Abstractions (e.g., x86 architecture)	4
3	Virtual to Physical Address Translation	3
4	Processes and Memory	3
5	Process Creation	1
6	Interrupts and Exceptions	2
7	Context-Switching	1
8	Process Scheduling	1
9	Multiprocessors and Locking	2
10	More Synchronization Primitives: Semaphores, Condition Variables, Barriers, etc.	4
11	Virtual Memory Swapping and Page Replacement	3
12	Files and Disk I/O	4
13	Crash Recovery and Logging	4

14	Lock-free Synchronization: CAS, RCU, etc.	3
15	Security and Access Control	2
16	Review	1
COURSE TOTAL *(14 times 'L')*		42

16.	**Brief description of tutorial activities**

	N.A.

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	Initializing PC Hardware : Device Handling, Stack, Interrupt/Exception Handling, Privilege Levels and Protected Control Flow Transfer.	6
2	Implementing User-level Processes by implementing system calls, command-line argument passing, address spaces, and protection mechanisms (e.g., based on Pintos user programs assignment).	7
3	Implementing Virtual Memory Mechanisms: Swapping to disk, memory-mapped files, copy-on-write optimizations, page replacement algorithms, stack growth (e.g., based on Pintos virtual memory assignment).	7
4	Implementing File System Mechanisms: directory structure, buffer cache, fine-grained filesystem synchronization, write behind and read ahead optimizations for the buffer cache (e.g., based on Pintos filesystem assignment)	8
COURSE TOTAL *(14 times 'P')*		28

18.	**Suggested texts and reference materials** Cox, Kaashoek, and Morris. Xv6: a teaching operating system, book draft as of Aug. 28, 2012.

	Silberschatz, Galvin, and Gagne. Operating System Concepts (Eighth edition). John Wiley & Sons, Inc., 2008. ISBN 0-470-12872-0.

1. Project specific

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	GNU/Linux, pintos, bochs, qemu : all free and open source software
19.2	Hardware	Commodity desktop, laptop and server machines
19.3	Teaching aides (videos, etc.)	Course webpage
19.4	Laboratory	Enough desktop terminals with all required software installed on them.
19.5	Equipment	None
19.6	Classroom infrastructure	Data projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	30%
20.2	Open-ended problems	
20.3	Project-type activity	50%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Principles of Artificial Intelligence
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 671
6.	**Status** (*category* for *program*)	Bridge Course for MTech (MCS),

7.	**Pre-requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	30% Overlap with COL 720 (Advanced Artificial Intelligence) 90% Overlap with COL 333 (Artificial Intelligence)
8.2	Overlap with any UG/PG course of other Dept./Centre	

9.	**Not allowed for** (indicate program names)	Students done equivalent course are not allowed

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	Saroj Kaushik, Mausam, Parag Singla, KK Biswas
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course introduces the ideas in the field of artificial intelligence. It is tailored to students who have not had any exposure of the subject. Since AI is quite a broad field, this course gives an overview, both of traditional and modern AI, and prepares a student for advanced elective courses.

14.	**Course contents (about 100 words) (Include laboratory/design activities):** Problem solving, search techniques, control strategies, game playing (minimax), reasoning, knowledge representation through predicate logic, rule based systems, semantics nets, frames, conceptual dependency. Planning. Handling uncertainty: probability theory, Bayesian Networks, Dempster-Shafer theory, Fuzzy logic, Learning through Neural nets - Back propagation, radial basis functions, Neural computational models - Hopfield Nets, Boltzman machines. PROLOG programming. Expert Systems, Soft computing, introduction to natural language processing.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Course Outline; Philosophy of Artificial Intelligence	3
2	Uninformed Search, Heuristic Search and Local Search	6
3	Constraint Satisfaction	3
4	Game Playing	3
5	Automated Classical Planning	3
6	Knowledge Representation: Logic and Satisfiability	4
7	Uncertainty in AI: Fuzzy Logic and Bayesian networks	6
8	Decision theory and Markov decision processes	2
9	Agents and Reinforcement learning	3
10	Machine learning and Neural networks	4
11	Soft computing	2
12	Intro to Natural language processing	2
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

N.A.

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	Modeling problems as search	5
2	Modeling problems as satisfiability	6
3	Implementing inference and learning for Bayesian nets	6
4	Modeling problems as an MDP	6
5	Use of Weka for basic machine leanring	5
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Stuart Russell & Peter Norvig, “Artificial Intelligence: A Modern Approach”,
1. Prentice-Hall, Third Edition (2009)
2. Saroj Kaushik, “Artificial Intelligence”, Cengage Learning, 2011.
3. Tom Mitchell. “Machine Learning” First Edition, McGraw-Hill, 1997.
4. Online Reading Material.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download (e.g. Weka machine learning repository, UCI datasets). Matlab may be used for couple of assignments.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	10%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Computer Networks
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 672
6.	**Status** (*category* for *program*)	Bridge Course for MTech (MCS),

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	COL 334
8.2	Overlap with any UG/PG course of other Dept./Centre	EE Lab on Computer Networks

9.	**Not allowed for** (indicate program names)	Students done equivalent course are not allowed

10.	**Frequency of offering**	Every other semester

11.	**Faculty who will teach the course**	Vinay Ribeiro, Huzur Saran, Aaditeshwar Seth, B.N. Jain
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: Will expose students to the fundamentals of computer networks, including concepts of layering, packet switching, protocols, connection orienting and statefulness in protocols, medium access control, naming and addressing, routing, reliable communication, and introduce students to network programming using the socket API, basic network configuration, packet sniffing, and Internet architecture.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Students will be exposed to common network algorithms and protocols, including physical layer modulation (analog AM/FM, digital ASK/FSK/PSK), encoding (NRZ, Manchester, 4B/5B), link layer framing, error control, medium access control (TDMA, FDMA, CSMA/CA, CSMA/CD), bridging, SDN, addressing (IPv4/v6), name resolution (DNS), routing (DV, LS, protocols RIP, OSPF, BGP), transport protocols (TCP), congestion avoidance (window based AIMD), and application design models

	(clientserver,P2P, functioning of HTTP, SMTP, IMAP). Programming assignments will be designed to test network application design concepts, protocol design towards developing error detection and correction methods, efficient network utilization, and familiarization with basic tools such as ping, traceroute, wireshark.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Introduction	2
2	Fundamentals: Protocols, connection oriented vs connection less, layers	3
3	Physical layer modulation and encoding	3
4	Link layer framing, error control, medium access control protocols, bridging, introduction to SDNs, wireless	9
5	IP protocol, IP addressing, IPv4/v6	2
6	Routing: Distance vector, link state, protocols, count-toinfinity	8
7	TCP, congestion avoidance, flow control	9
8	Application design	3
9	Name resolution, DNS	1
10	Introduction to Network Security: SSL, IPSec, TLS	2
COURSE TOTAL ***(14 times 'L')***		42

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	**Experiment description**	**No. of hours**
1	Familiarization with ping, traceroute, Internet architecture	4
2	Wireshark to understand link layer, IP, TCP, application (HTTP, DNS)	4
3	Socket programming to design application	10
4	Algorithms for error control and correction	5
5	Algorithms for network utilization	5
COURSE TOTAL ***(14 times 'P')***		28

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Kurose and Ross (book), Computer Networking: A Top-down Approach, 6th edition, 2012. [core text]
2. Peterson and Davie (book), Computer Networks: A Systems Approach, 5th edition, 2011. [reference]
3. S. Keshav. An Engineering Approach to Computer Networks, 1st edition, 1997. [reference]

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N/A
19.4	Laboratory	N/A
19.5	Equipment	N/A
19.6	Classroom infrastructure	LCD Projector
19.7	Site visits	N/A2

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: The course is designed to be a mix of theory as well as hands on experience. A background in Basic Probability is required.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Data Structures and Algorithms
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL702
6.	**Status** (*category* for *program*)	PC for MTech (MCS), PE for CSY/SIY/ANZ/CSZ

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	COL106
8.2	Overlap with any UG/PG course of other Dept./Centre	Data Structure of Maths

9.	**Not allowed for** (indicate program names)	CS1, CS5

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	Sandeep Sen, Amitabha Bagchi, Naveen Garg, Amit Kumar, Ragesh Jaiswal, Subodh Kumar, Saroj Kaushik
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The students will learn about various data-structures and related algorithms with emphasis on time and space complexity. The course will involve extensive programming to implement and apply the data structures learnt.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Review of basic data structures and their realization in object oriented Environments. Dynamic Data structures: 2-3 trees, Redblack trees, binary heaps, binomial and Fibonacci heaps, Skip lists, Universal Hashing. Data structures for maintaining ranges, intervals and disjoint sets with applications. B-trees. Tries and suffix trees. Priority queues and binary heaps. Sorting: merge, quick, radix, selection and heap sort, Graphs: Breadth first search and connected components. Depth first search in directed and undirected graphs. Disjkra's algorithm, Directed acyclic graphs and topological sort. Some geometric data-structures. Basic algorithmic techniques like dynamic programming and divide-and-conquer. Sorting algorithms with analysis, integer sorting, selection. Graph algorithms like DFS with applications, MSTs and shortest paths.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to data-structures and asymptotic running time analysis	2
2	Object oriented concepts and abstract data-types	1
3	Stacks, queues and applications	3
4	Linked lists, doubly linked lists, sequences	2
5	Dictionaries and implementation using skip lists	2
6	Hashing: collision resolution techniques	2
7	Trees and traversal algorithms	2
8	Binary search trees: insertion, deletion and average case analysis	4
9	AVL Trees: insertion and deletion	3
10	2-4 trees, Red-Black trees	2
11	B-trees	2

12	Tries, suffix trees and pattern matching	3
13	Priority queues and binary heaps	2
14	Sorting algorithms	4
15	Graphs: introduction, BFS, DFS, topological sort	4
16	Shortest paths – Dijkstra's algorithm	2
17	Geometric data-structures	2
COURSE TOTAL ***(14 times 'L')***		42

16.	**Brief description of tutorial activities: NA**

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Stacks/queues with objected oriented features	4
2	Trees: binary, AVL, Red-black, B-trees	6
3	Hashing	4
4	Priority queues	2
5	Tries and suffix trees	2
6	Graphs: DFS, BFS	4
7	Shortest path algorithm implementation and applications	4
8	Geometric data-structures and applications	2
COURSE TOTAL ***(14 times 'P')***		28

18.	**Suggested texts and reference materials**

1. Michael T. Goodrich and Roberto Tamassia, Data-structures and Algorithms in JAVA, 5th edition, John Wiley and Sons Co., 2010.
2. Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, Introduction to Algorithms, MIT (2010).

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Java, C++ development environments
19.2	Hardware	Access to computers
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	Computers with java/C++ development environments
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	10%
20.2	Open-ended problems	N.A.
20.3	Project-type activity	10%
20.4	Open-ended laboratory work	20%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Logic for Computer Science
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	CSL 703
6.	**Status** (*category* for *program*)	Soft PC for MTech (MCS), PC for CS5 (Dual), PE for CSY/SIY/ANZ/CSZ DE for CS1

7.	**Pre Requisites** (course no./title)	CSL201/CSL160 Data Structures

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	10% with CSL765
8.2	Overlap with any UG/PG course of other Dept./Centre	60% with MAL607
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	Sanjiva Prasad, S. Arun-Kumar
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: The course provides a rigorous formal introduction to mathematical logic – its syntax, semantics and proof techniques – and its applications in computer science. .

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Review of the principle of mathematical induction; the principle of structural induction; review of Boolean algebras; Syntax of propositional formulas; Truth and the semantics of propositional logic; Notions of satisfiability, validity, inconsistency; Deduction systems for propositional logic; Completeness of deduction systems; First order logic (FOL) ; Proof theory for FOL; introduction to model theory; completeness and compactness theorems; First order theories. Programming exercises will include representation and evaluation; conversion to normal-forms; tautology checking; proof normalization; resolution; unification; Solemnization, conversion to Horn-clauses; binary-decision diagrams. Decidability, undecidability and complexity results.

	Introduction to formal methods, temporal/modal logics.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Mathematical foundations and induction principles	3
2	Propositional logic: syntax and semantics	3
3	Proof techniques for propositional logic	5
4	Logical equivalence, consequence, canonical forms	3
5	First order logic (FOL); syntax and semantics	5
6	Models, satisfiability, validity and unsatisfiability	4
7	Proof systems for FOL	6
8	Soundness, completeness and compactness theorems for FOL	5
9	Decidability and undecidability	3
10	Foundations of formal methods	2
11	Temporal/Modal logics	3
COURSE TOTAL (14 times 'L')		42

16.	**Brief description of tutorial activities** - NA

17.	**Brief description of laboratory activities**:

Module no.	Experiment description	No. of hours
1	Evaluation of propositions and conversion to normal forms	4
2	Binary-decision diagrams	4
3	Proof normalization	6
4	Tableaux methods, resolution	8
5	Model-checking for LTL/CTL	6
COURSE TOTAL (14 times 'P')		28

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. H. D. Ebbinghaus, J. Flum, and W. Thomas. Mathematical Logic. Springer-Verlag, New York, USA, 1994.
2. H. B. Enderton. A Mathematical Introduction to Logic. Elsevier India, New Delhi, India, 2001.
3. M. Fitting. First-Order Logic and Automated Theorem Proving. Springer-Verlag, New York,

USA, 1990.
4. M. Huth and M. Ryan. Logic in Computer Science: Modelling and Reasoning about Systems. Cambridge University Press, Cambridge, UK, 2000.
5. E. Mendelson. Introduction to Mathematical Logic. D. Van Nostrand Co. Inc., Princeton, New Jersey, USA, 1963.
6. Anil Nerode and R. Shore. Logic for Applications. Springer-Verlag, New York, USA, 1993.
7. R. M. Smullyan. First-Order Logic. Springer-Verlag, Berlin, Germany, 1968.☐

19.	Resources required for the course (itemized & student access requirements, if any)

19.1	Software	SML/OCAML and Prolog
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	Design content of the course (Percent of student time with examples, if possible)

20.1	Design-type problems	15%
20.2	Open-ended problems	20%
20.3	Project-type activity	25%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Theory of Computation and Complexity
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 705
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 352 (Introduction to Automata and theory of Computation) or equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	Frequency of offering	Every semester

11.	**Faculty who will teach the course**	Ragesh Jaiswal, Sandeep Sen, Amit Kumar
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)***:** The objective of this course is to familiarise students with computability and complexity using a universal computation model like Turing Machine. Starting with undecidability, the course explores many structural complexity classes like P, NP, PSPACE, #P, PH, and Randomized Complexity as well as their relationships .

14.	Course contents (about 100 words) (Include laboratory/design activities): Review of Automata Theory, Turing Machines and Universal Turing Machines. Computability & Undecidability, Rice's theorem. Computational Complexity: Time and Space hierarchy, Gap theorem, Classes P and NP, Cook-Levin theorem, PSPACE completeness, NL completeness, Immerman-Szelepcseyi theorem, Alternation, Polynomial hierarchy, Permanent and #P completeness, Oracle machines, Baker-Gill-Solovay theorem, Randomized Turing Machine, RP, ZPP, BPP, Adleman's theorem, Yao's minmax theorem, Derandomization,

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Turing Machine, Recursive and Recursively enumerable languages	3
2	Goedel numbering, Universal TM, Undecidability	3
3	Rice's theorem, Post Correspondence problem	3
4	Time and Space models, linear speed-up	3
5	Savitch's theorem, Time and Space hierarchy, Borodin's gap theorem	4
6	Class P and NP, log space, Cook-Levin theorem	6
7	PSPACE completeness	3
8	NL Completeness, Immerman's theorem	4
9	Alternation, Polynomial hierarchy, Oracle machines, #P Completeness	6
10	Randomized Turing Machines and Complexity Classes ZPP, RP, BPP	5
11	Relationship between BPP and P	2
COURSE TOTAL (14 times 'L')		42

16.	Brief description of tutorial activities - NA

17.	Brief description of laboratory activities: NA

Module no.	**Experiment description**	**No. of hours**
1		
2		
3		
4		
COURSE TOTAL *(14 times 'P')*		

18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. J. Hopcroft and J. Ulman, Introduction to Automata Theory, Languages and Computation, Narosa Pub, 1987
2. C. Papadimitrou, Computational Complexity, Addson Wesley 1995.
3. S. Arora and B. Barak, Computational Complexity A modern Approach, Cambridge 2009.
4. D. Kozen, Theory of Computation, Springer 2006.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	.
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: Knowledge of Discrete Mathematics and Basic probability will be expected

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Architecture of High Performance Computers
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 718
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL216,COL373

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	NA
10.	**Frequency of offering**	2nd sem
11.	**Faculty who will teach the courses : Anshul Kumar, Kolin Paul, Smruti Sarangi**	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is designed as a followup of a basic course in Computer Architecture (CSL211 or equivalent). The objective is to discuss the advanced architectural concepts which improve the performance of computer systems. Instruction level as well as system level parallelism are considered
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Classification of parallel computing structures; Instruction level parallelism - static and dynamic pipelining, improving branch performance, superscalar and VLIW processors; High performance memory system; Shared memory multiprocessors and cache coherence; Multiprocessor interconnection networks; Performance modelling; Issues in programming multiprocessors; Data parallel architectures

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	General principles of performance enhancement	4
2	Instruction level parallel architectures	10
3	High performance memory systems	7
4	Shared memory multiprocessors	6
5	Multiprocessor interconnection networks	5
6	Performance modelling	3
7	Programming multiprocessor	3
8	Data parallel and multi-threading thread architectures	4
9		
10		
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Use of performance evaluation/simulation tools to compare architectures.	14
2	Implementing models of architectural features and studying their performance	14
	COURSE TOTAL *(14 times 'P')*	28

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Sima D., Fountain T., Kacsuk P., Advanced Computer Architectures : A Design Space Approach", Addison Wesley, 1997.
2. Flynn M.J., , "Computer Architecture : Pipelined and Parallel Processor Design", Narosa Publishing House.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	

20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Synthesis of Digital Systems
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 719
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL316

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	none
10.	**Frequency of offering**	☐ Every sem ☒ 1st sem ☐ 2nd sem ☐ Alternate year

11.	Faculty who will teach the course 1. P.R. Panda 2. M. Balakrishnan 3. Anshul Kumar	
12.	**Will the course require any visiting faculty?**	No
13.	**Course objective** *(about 50 words)*: The course covers the conceptual and practical issues in the automatic synthesis of VLSI circuits. Hardware description languages are introduced. The student is taken through different levels of hardware abstraction, and the issues involved in the automation process called synthesis, which helps transition from one level to the next.	
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*:	

	After a basic overview of the VLSI design flow, hardware modelling principles and hardware description using the VHDL language are covered. This is followed by a study of the major steps involved in behavioural synthesis: scheduling, allocation, and binding. This is followed by register-transfer level synthesis, which includes retiming and Finite State Machine encoding. Logic synthesis, consisting of combinational logic optimisation and technology mapping, is covered next. Popular chip architectures - standard cells and FPGA are introduced. The course concludes with a brief overview of layout synthesis topics: placement and routing.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction: VLSI design flow	2
2	Hardware Description Languages	6
3	Behavioural synthesis	8
4	. Retiming and FSM synthesis	4
5	Logic synthesis	8
6	Technology mapping	2
7	Timing analysis	2
8	Field Programmable Gate Array	4
9	Placement and Routing	4
10	Other: Verificatiion, Testing, Testability	2
11		
12		
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Using graph package and compiler infrastructure	3
2	VHDL modelling and simulation	4
3	Scheduljing	7
4	FSM generation	7
5	Standard cell placement and routing	7
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		28

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Micheli G., Synthesis and Optimization of Digital Circuits, McGraw-Hill Intl.,1994

2. Michel P, Lauther U, Duzy P,, The Synthesis Approach to Digital System Design, Kluwer Academic Publishers, 1992
3. Gajski D., Dutt N., Wu A., Lin S., High-Level Synthesis: Introduction to Chip and System Design, Kluwer Academic Publishsers, 1992
4. Navabi Z., VHDL: Analysis and Modelling of Digital Systems, McGraw-Hill Intll., 1998

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	SUIF Compiler infrastructure (open source), LEDA graph package (open source), GHDL – VHDL Simulator (open source)
19.2	Hardware	PC with Linux
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	Sufficient number of PCs
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	80%
20.4	Open-ended laboratory work	20%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	SYSTEM LEVEL DESIGN AND MODELLING
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL812
6.	**Status** (*category* for *program*)	PE for **CO**

7.	**Pre-requisites** (course no./title)	COL719

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☒ 2nd sem ☐ Alternate year
11.	**Faculty who will teach the course**	1. M Balakrishnan, P R Panda
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To understand the principles of modelling and design of complex embedded systems with both hardware and software components.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Embedded systems and system-level design, models of computation, specification languages, hardware/software co-design, system partitioning, application specific processors and memory, low power design.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to embedded systems	3
2	Models of computation	4
3	Specification languages	3
4	Behavioural synthesis review	3
5	Hardware/software codesign	5
6	Application specific processors	4
7	Application specific memory	4
8	Low power design	4
9	Special topics and recent developments	10
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Modelling and simulation of a reasonably complex embedded system application.	
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Michel, P., Lauther, U., and Duzy, P. (eds.), The synthesis approach to digital systems, Kluwer Academic Publishers, Norwell, 1992.
2. Gajski, D. D., Vahid, F., Narayan, S., and Gong, J., Specification and Design of Embedded Systems, Prentice Hall, New Jersey, 1994
3. Lee, E. A., and Parks, T. M., Dataflow Process Networks, Proceedings of the IEEE, May 1995, pg 773-801
4. SystemC Version 2.0 User's Guide
5. Smith, M. J. S., Application -Specific Integrated Circuits, Pearson Education,1997 (Chapter 15, pg 824-838)
6. Jain, M. K., Balakrishnan, M., and Kumar, A., ASIP Design Methodologies: Survey and Issues, International Conference on VLSI Design, 2001

7. Panda, P. R., Dutt, N. D., and Nicolau, A., Memory issues in embedded systems-on-chip: optimizations and exploration, Kluwer Academic Press, 1999.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	RECONFIGURABLE COMPUTING
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL821
6.	**Status** (*category* for *program*)	PE for **CO**

7.	**Pre-requisites** (course no./title)	COL719

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐Every sem ☒1st sem ☐2nd sem ☒Alternate year
11.	**Faculty who will teach the course**	Kolin Paul, Anshul Kumar
12.	**Will the course require any visiting faculty?**	No

13. Course objective *(about 50 words)*:

The objective of this course is to familiarize the students with the current research on various aspects of reconfigurable computing, including reconfigurable devices, architecture and compilers. The course will involve a fair amount of critically analyzing a digest of papers. There will also be a semester long implementation project.

14. Course contents *(about 100 words) (Include laboratory/design activities)*:

FPGA architectures, CAD for FPGAs: overview, LUT mapping, timing analysis, placement and routing, Reconfigurable devices - from fine-grained to coarse-grained devices, Reconfiguration modes and multi-context devices, Dynamic reconfiguration, Compilation from high level languages, System level design

	for reconfigurable systems: heuristic temporal partitioning and ILP-based temporal partitioning, Behavioral synthesis, Reconfigurable example systems' tool chains

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to Reconfigurable Computing	1
2	Designing for FPGA	2
3	Logic Cells, Tech Mapping and Placement	3
4	CAD of FPGAs - a) the Xilinx tool chain, b) building systems with HandelC	3
5	Systolic architectures and applications	6
6	Systolic loop transformations - a) dependence analysis, b) scheduling, c) localization	6
7	Software pipelining	3
8	Partial reconfiguration, multi-context architectures	3
9	Runtime reconfigurable architectures, Piperench	3
10	Behavioral synthesis	6
11	Example reconfigurable systems' tool chains - COMPAAN (with KPNs), StreamsC (with CSP), MMAlpha	6
12		
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	There will be a semester long project as a part of the course	
2		
3		
4		
5		
6		
7		
8		
9		
10		
	COURSE TOTAL *(14 times 'P')*	

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

Scott Hauck and Andre DeHon, *Reconfigurable Computing: The Theory and Practice of FPGA-Based Computation (Systems on Silicon)* Morgan Kaufmann 2007

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	SPECIAL TOPICS IN HARDWARE SYSTEMS
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL861
6.	**Status** (*category* for *program*)	PE for **CO**

7.	**Pre-requisites** (course no./title)	E.C. 120

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year
11.	**Faculty who will teach the course** Anshul Kumar, M. Balakrishnan, Preeti R Panda	
12.	**Will the course require any visiting faculty?**	No
13.	**Course objective** *(about 50 words)*:	
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Under this topic one of the following areas will be covered: Fault Detection and Diagnosability. Special Architectures. Design Automation Issues. Computer Arithmetic, VLSI	

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
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9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	INDEPENDENT STUDY
3.	**L-T-P structure**	0-3-0
4.	**Credits**	3
5.	**Course number**	COS310
6.	**Status** (*category* for *program*)	DE for **CS,** DE for **CO**

7.	**Pre-requisites** (course no./title)	EC 60

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	non-CS students
10.	**Frequency of offering**	☒Every sem ☐1st sem ☐2nd sem ☐Alternate year
11.	Faculty who will teach the courseAll	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Research oriented activities or study of subjects outside regular course offerings under the guidance of a faculty member. Prior to registration, a detailed plan of work should be submitted by the student to the Head of the Department for approval
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Research oriented activities or study of subjects outside regular course offerings under the guidance of a faculty member. Prior to registration, a detailed plan of work should be submitted by the student to the Head of the Department for approval

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	SPECIAL MODULE IN HARDWARE SYSTEMS
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV881
6.	**Status** (*category* for *program*)	PE for **CO**

7.	**Pre-requisites** (course no./title)	Permission of Instructor

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐Every sem ☐1st sem ☐2nd sem ☒Alternate year
11.	Faculty who will teach the course	
12.	**Will the course require any visiting faculty?**	

13. **Course objective** *(about 50 words)*:
To provide insight into current research problems in this area

14. **Course contents** *(about 100 words) (Include laboratory/design activities)*:
Special module that focuses on special topics and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
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7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Introduction to Compressed Sensing
3	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 722
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No
8.3	Supersedes any existing course	No

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Every year

11.	**Faculty who will teach the course**	Rahul Garg
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Compressed sensing is an emerging technique in machine learning with applications in diverse areas. The course will present the basic theory of compressed sensing and sparse reconstruction from under-sampled data. It will also cover recent ideas in modern convex optimization allowing rapid signal recovery or parameter estimation and give a flavor of applications that can potentially benefit from compressive sensing techniques. The overarching theme is to expose students to a novel active field of research and give them the tools to make theoretical and/or practical contributions.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Sparsity, L1 minimization, Sparse regression, deterministic and probabilistic approaches to compressed sensing, restricted isometry property and its application in sparse recovery, robustness in the presence of noise, algorithms for compressed sensing. Applications in magnetic resonance imaging (MRI), applications in analog-to-digital conversion, low-rank matrix recovery, applications in image reconstruction.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Recap of regression and regularization	6
2	L1 penalty and its properties	6
3	Sparse regression	3
4	Restricted isometry property	3
5	Theoretical results on sparse recovery using L1 penalty	3
6	Algorithms for sparse recovery	3
7	Application in MRI	6
8	Applications in analog-to-digital conversion	3
9	Applications in low-rank matrix recovery	3
10	Applications in other image reconstruction problems	6
COURSE TOTAL ***(14 times 'L')***		42

16.	**Brief description of tutorial activities:** NA

17.	**Brief description of laboratory activities:** NA

Module no.	**Experiment description**	**No. of hours**
COURSE TOTAL ***(14 times 'P')***		

18.	**Suggested texts and reference materials**

1. Probability and Random Processes by G. Grimmett and D. Stirzaker, 3rd. ed., Oxford University Press
2. Random Matrices by M. Mehta, 3rd ed., New York: Academic Press
3. Function Estimation and Gaussian Sequence Models by I. Johnstone (PDF), Chapter 13 (PDF)
4. All of Statistics by L. Wasserman, Springer
5. Numerical Linear Algebra by LLoyd N. Trefethen and David Bau, III, SIAM
6. Convex Optimization by S. Boyd and L. Vandenberghe, Cambridge University Press
7. Introductory Lectures on Convex Optimization: A Basic Course by Y. Nesterov, Kluwer Academic Publisher
8. A Wavelet Tour of Signal Processing by S. Mallat, 3rd ed., Academic Press
9. Discrete Time Signal Processing by A. Oppenheim and R. Schafer, Prentice Hall

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	None
19.2	Hardware	Access to computers
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	10%
20.2	Open-ended problems	15%
20.3	Project-type activity	50%
20.4	Open-ended laboratory work	15%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Department of Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Computer Networks
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 724
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL374 Computer Networks or equivalent.

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	N/A
8.2	Overlap with any UG/PG course of another Dept./Centre	N/A
8.3	Supersedes any existing course	COL858

9.	**Not allowed for** (indicate program names)	N/A

10.	**Frequency of offering**	Every alternate semester

11.	**Faculty who will teach the course** 1. Huzur Saran, 2. V. Ribeiro, 3. Aaditeshwar Seth 2.	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words; motivation and aims of this course)***:** The course will expose students to advanced concepts such as wireless MAC, BGP routing, MPLS, QOS scheduling and flow control, TCP variants, etc, with the objective of taking them into further details of computer networks. Assignments and projects will focus on teaching research methods such as simulations and performance evaluation.
14.	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)***:** Review of the Internet architecture, layering; wired and wireless MAC; intra- and inter-domain Internet routing, BGP, MPLS, MANETs; error control and reliable delivery, ARQ, FEC, TCP; congestion and flow control; QoS, scheduling; mobility, mobile IP, TCP and MAC interactions, session persistence; multicast; Internet topology, economic models of ISPs/CDNs/content providers; future directions

15.	**Lecture Outline** *(with topics and number of lectures)*	
Module no.	**Topic**	**No. of hours**
1	Internet architecture recap and principles	4
2	Error & Flow control; TCP	6
3	Routing (BGP, Policy Routing, MANET, Cost functions, Dynamics)	6
4	Network performance; flow control; scheduling & active queue mgmt	6
5	IP Mobility & IPv6	2
6	Wireless MAC, RLP and challenges	4
7	Internet topology and traffic characteristics – measurements & models:	4
8	MPLS & Traffic Engineering	2
9	Overlay Networks & Content Distribution	4
10	Future directions of the Internet architecture	4
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities - NA

17.	**Brief description of laboratory activities**	
Module no.	**Experiment description**	**No. of hours**
1	Lab Exercise one – Bandwidth/latency measurements off distributed Internet nodes (PlanetLab)	4
2	Lab Exercise two – TCP performance in wireless on a local testbed	6
3	Lab Exercise three – Multicast Overlay Simulation with Mobility	8
4	Lab Project	10
8		
	COURSE TOTAL *(14 times 'P')*	28

18. Suggested texts and reference materials

1. Keshav, S. *An Engineering Approach to Computer Networks*
2. Shivkumar, *Network Architecture: Principles, Guidelines*, RPI 2006.
3. Peterson and Davie (book), *Computer Networks: A Systems Approach*
4. Relevant papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	Network Simulation Software
19.2	Hardware	Lab computing resources and Routers (available)
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	Proxy quota to run Internet measurements
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Numerical Algorithms
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 726
6.	**Status** (*category* for *program*)	Soft PC for MTech (MCS), Dual(CS), PE for MTech (MCS), CSY/SIY/ANZ/CSZ DE for CS1

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	MAL 230 (80% overlap)
8.3	Supersedes any existing course	CSL 361

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	Subhashis Bannerjee, Amit Kumar
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*:

	This course is intended to give students tools for understanding sources of errors in computation and analyzing numerical stability of algorithms. They will also learn various tools in numerical linear algebra for solving problems arising in regression, estimation and optimization.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Number representation, fundamentals of error analysis, conditioning, stability, polynomials and root finding, interpolation, singular value decomposition and its applications, QR factorization, condition number, least squares and regression, Gaussian elimination, eigenvalue computations and applications, iterative methods, linear programming, elements of convex optimization including steepest descent,conjugate gradient, Newton's method.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Number representation and rounding errors	2
2	Conditioning and stability	2
3	Polynomials and root finding	4
4	Interpolation	2
5	Matrix norms and singular value decomposition	4
6	QR factorization, Householder, Givens	4
7	Least squares, regression	4
8	Guassian elimination and its numerical stability	2
9	Eigenvalues and applications	4
10	Power method and its convergence	2
11	Iterative methods	2
12	Linear programming and simplex algorithm	4
13	Convex programming and applications	2
14	Steepest descent, conjugate gradient	4
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities -** NA

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Introduction to matlab and matrix computations	4
2	Singular Value Decomposition and applications	4
3	Least squares and regression	4
4	Applications of eignevalues	6
5	Linear programming	4
6	Convex optimization	6
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Lloyd N. Trefethen and David Bau III, Numerical Linear Algebra, 1st edition, Society for Industrial and Applied Mathematics, 1997
2. Michael Heath, Scientific Computing: An introductory survey, 2nd edition, Tata McGraw-Hill Publishing Company, 2011.
3. Gene H. Golub, Charles van Loan, Matrix Computations, 3rd edition, Johns Hopkins University Press, 1996.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Matlab
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	Computers with matlab

19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	10%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Compiler Design
3.	**L-T-P structure**	3-0-3
4.	**Credits**	4.5
5.	**Course number**	COL 728
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 226 (Programming Languages) COL 211 (Computer Architecture)

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	15% overlap with CSL 729 (new course)
8.2	Overlap with any UG/PG course of other Dept./Centre	EEL702
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	DE for BTech and Dual (CS) PE for MTech, Dual (CS)

10.	**Frequency of offering**	1stsem

11.	**Faculty who will teach the course**	S. Arun-Kumar, SoravBansal, PreetiRanjan Panda, Sanjiva Prasad
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This objective of this course is to introduce the student to theoretical and practical aspects of the implementation of a compiler: the front end, optimization and code generation

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Compilers and translators; lexical and syntactic analysis, top-down and bottom up parsing techniques; internal form of source programs; semantic analysis, symbol tables, error detection and recovery, code generation and optimization. Type checking and static analysis. Static analysis formulated as fixpoint of simultaneous semantic equations. Data flow. Abstract interpretation. Correctness issues in code optimizations. Algorithms and implementation techniques for type-checking, code generation and optimization. Students will design and implement translators, static analysis, type-checking and optimization. This is a praxis-based course. Students will use a variety of software tools and techniques in implementing a complete compiler.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Compilers, interpreters and translators	3
2	Lexical Analysis	3
3	Parsing, theory and practice	6
4	Abstract syntax and internal representations	2
5	Type checking	4
6	Static Analysis fundamentals and theory	4
7	Static analyses: Data and/or control flow	4
8	Abstract interpretation	2
9	Code generation	6
10	Optimizations of various kinds	6
11	Architecture specific optimizations	2
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities -** NA

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Lexical Analysis using lex/flex	4
2	Parsing using Yacc/Bison	14
3	Static Analyses	12
4	Code Generation and optimizations	12
COURSE TOTAL *(14 times 'P')*		**42**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Andrew Appel. Modern Compiler Implementation in ML. Cambridge Univ Press, 1998. *(*Also *in C/Java).*
2. A. Aho, R. Sethi, J. Ullman. Compilers: Principles, Techniques, and Tools. Addisson-Wesley, 2006.
3. S. S. Muchnick, Advanced Compiler Design and Implementation, Morgan Kaufman, 1997
4. R. Allen and K. Kennedy, Optimizing Compilers for Modern Architectures, Morgan Kaufman, 2001.
5. F. Nielson, H. R. Nielson, C. Hankin. Principles of Program Analysis. Springer. 2005.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	ML/C++/Java compiler
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.

19.6	Classroom infrastructure	White board, Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	30%
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	30%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Compiler Optimisation
3.	**L-T-P structure**	3-0-3
4.	**Credits**	4.5
5.	**Course number**	CSL 729
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 226 (Programming Languages) COL 211 (Computer Architecture)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	15% overlap with CSL 728 (Compiler Design)
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Alternate Semesters

11.	**Faculty who will teach the course**	P. R. Panda, Sorav Bansal, Kolin Paul
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: This objective of this course is to introduce the student to theoretical and practical

	aspects of optimisation techniques employed in modern compilers.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Program representation – symbol table, abstract syntax tree; Control flow analysis; Data flow analysis; Static single assignment; Def-use and Use-def chains; Early optimisations – constant folding, algebraic simplifications, value numbering, copy propagation, constant propagation; Redundancy Elimination – dead code elimination, loop invariant code motion, common sub-expression elimination; Register Allocation; Scheduling – branch delay slot scheduling, list scheduling, trace scheduling, software pipelining; Optimizing for memory hierarchy – code placement, scalar replacement of arrays, register pipelining; Loop transformations – loop fission, loop fusion, loop permutation, loop unrolling, loop tiling; Function inlining and tail recursion; Dependence analysis; Just-in-time compilation; Garbage collection. Laboratory component would involve getting familiar with internal representations of compilers; profiling and performance evaluation; and the design and implementation of novel compiler optimisations.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Program representation	3
2	Control flow analysis	2
3	Data flow analysis	4
4	Static single assignment	2
5	Early optimisations: constant folding, algebraic simplifications, value numbering, copy propagation, constant propagation	4
6	Redundancy Elimination – dead code elimination, loop invariant code motion, common sub-expression elimination	3
7	Register Allocation	4
8	Scheduling – branch delay slot scheduling, list scheduling, trace scheduling, software pipelining	4
9	Optimising for memory hierarchy – code placement, scalar replacement of arrays, register pipelining	4
10	Loop transformations – loop fission, loop fusion, loop permutation, loop unrolling, loop tiling	4

11	Function inlining and tail recursion	2
12	Dependence analysis	2
13	Other topics: Just-in-time compilation, Garbage collection, Processor-specific optimisation	4
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

N.A.

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Getting familiar with internal representation	6
2	Design of a compiler optimisation algorithm	10
3	Software implementation of optimisation pass	20
4	Testing of implementation, Profiling	6
COURSE TOTAL ***(14 times 'P')***		**42**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. S. S. Muchnick, Advanced Compiler Design and Implementation, Morgan Kaufman, 1997
2. R. Allen and K. Kennedy, Optimizing Compilers for Modern Architectures, Morgan Kaufman, 2001
3. J. Aycock, A brief history of just-in-time, ACM Computing Surveys, 35(2), 2003
4. M. Arnold, S. J. Fink, D. Grove, M. Hind, and P. F. Sweeney, A Survey of Adaptive

Optimization in Virtual Machines, Proc. of the IEEE, 93(2), 2005
5. Paul R. Wilson, Uniprocessor Garbage Collection Techniques. Proc. Intl. Workshop on Memory Management (IWMM), Springer-Verlag, 1992

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java compiler
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	White board, Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	30%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Parallel Programming
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL730
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL106, COL331

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	NIL
8.2	Overlap with any UG/PG course of other Dept./Centre	NIL

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either Sem.

11.	**Faculty who will teach the course**	Subodh Kumar, Rahul Garg, Sandeep Sen
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This is a course in intensive parallel programming. With the growing number of cores on a chip, programming them efficiently has become an indispensable. This is a hands-on course involving significant parallel programming on compute-clusters, multi-core CPUs and massive-core GPUs.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Parallel computer organization, Parallel performance analysis, Scalability, High level Parallel programming models and framework, Load distribution and scheduling, Throughput, Latency, Memory and Data Organizations, Inter-process communication and synchronization, Shared memory architecture, Memory consistency, Interconnection network and routing, Distributed memory architecture, Distributed shared memory, Parallel IO, Parallel graph algorithms, Parallel Algorithm techniques: Searching, Sorting, Prefix operations, Pointer Jumping, Divide-and-Conquer, Partitioning, Pipelining,

	Accelerated Cascading, Symmetry Breaking, Synchronization (Locked/Lock-free).

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction, History and current state of parallel computers	2
2	Parallel Architecture Classes, Communication networks, Routing	4
3	Scalability and Performance analysis	3
4	Parallel Task Design, Task graphs, Load Balancing, Scheduling	4
5	Programming frameworks like OpenMP, MPI, Map-reduce, Global Arrays	4
6	Parallel Algorithms and Techniques I: Prefix operations, Pointer Jumping, Divide-and-Conquer	4
7	Parallel Algorithms and Techniques II: Partitioning, Pipelining, Accelerated Cascading, Symmetry Breaking	4
8	CUDA, OpenCL, Accelerator programming frameworks	4
9	Concurrent Data Structures including Lock-free/Wait-free operation	4
10	Searching/Sorting	3
11	Parallel IO	3
12	Parallel graph algorithms	3
COURSE TOTAL ***(14 times 'L')***		42

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	Experiment description	No. of hours
1	To be re-designed each semester: Will include three exercised on training on many-core devices as well as shared memory and distributed memory architectures. The final exercise will be a project integrating all three types of environment, with emphasis on scalability (when sufficient computing resource is available).	3*8+4
2		

3		
4		
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. An Introduction to Parallel Algorithms by Joseph Jaja (Addison-Wesley Professional), 1992.
2. Introduction to Parallel Computing by Ananth Grama, George Karypis, Vipin Kumar and Anshul Gupta (Pearson), 2003.
3. Parallel Programaming in C with MPI and openMP by Michael J Quinn (McGraw Hill), 2003.
4. Programming Massively Parallel Processors: A Hands-on Approach by David B. Kirk, and Wen-mei W. Hwu (Morgan Kaufmann), 2010.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	OpenMP, MPI, Cuda
19.2	Hardware	Compute cluster with Cuda devices
19.3	Teaching aides (videos, etc.)	Slide projector
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Virtualization and Cloud Computing
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL732
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5
7.	**Pre-requisites** (course no./title)	COL331

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	-
8.2	Overlap with any UG/PG course of another Dept./Centre	-
8.3	Supersedes any existing course	-

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Alternate Semester

11.	**Faculty who will teach the course** 1. Dr Sorav Bansal **2.** Prof. Huzur Saran	
12.	**Will the course require any visiting faculty?**	NO

13.	**Course objective** *(about 50 words; motivation and aims of this course)*: The course introduces the students to state-of-the-art research in the fields of Machine Virtualization and Cloud Computing.

14.	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)*: Introduction to Virtualization and Cloud Computing, Binary Translation, Hardware Virtualization, Memory Resource Management in Virtual Machine Monitor, Application of Virtualization, Cloud-scale Data Management and Processing, I/O Virtualization

15.	**Lecture Outline** *(with topics and number of lectures)*	
Module no.	**Topic**	**No. of hours**
1	Introduction, Binary Translation, Hardware, Virtualization	6
2	Para-Virtualization	2
3	Memory Management in Virtual Machine Monitor	4
4	I/O Virtualization	7
5	Application of Virtualization	6
6	Distributed File System	4
7	Large Cloud-scale Data Management	4
8	Cloud-scale Data Processing	4
9	Cloud Pricing	2
10	Nested Virtualization	3
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16.	**Brief description of tutorial activities**
	None

17.	**Brief description of laboratory activities**	
Module no.	**Experiment description**	**No. of hours**
1	VMM Benchmarking	9
2	Binary Translation and Shadow Page Tables	10
3	Hardware and I/O Virtualization	9
	COURSE TOTAL *(14 times 'P')*	28

18.	**Suggested texts and reference materials** Research Papers (no suitable book). Some papers: 1. A Comparison of Software and Hardware Techniques for x86 Virtualization (Adams, Agesen, 2006) 2. Xen and the Art of Virtualization (Barham et. Al., 2003). 3. Memory Resource Management in VMware ESX Server (Waldspurger 2002). 4. Accelerating Two-Dimensional Page Walks for Virtualized Systems (Bhargava et. al., 2008). 5. Virtualizing I/O Devices on VMware Workstation's Hosted Virtual Machine Monitor (Sugerman et. al., 2001). 6. ELI: Bare-metal Performance for I/O virtualization (Gordon et. al., 2012) 7. The Google Filesystem (Ghemawat et. al., 2003) 8. Bigtable: A Distributed Storage System for Structured Data (Chang et. al., 2006) 9. MapReduce: Simplified Data Processing on Large Clusters (Dean et. al., 2004) 10. The Turtles Project: Design and Implementation of Nested Virtualization (Ben-Yehuda et. al., 2010) Other resources (for lab programming) 1. Linux/KVM (www.linux-kvm.org) 2. Binary translation based VMMs 3. Online cloud implementations like Amazon EC2, Google AppEngine, Microsoft Azure

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	KVM, Linux (all open source)
19.2	Hardware	X86-based computers
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	
20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	20%
20.2	Open-ended problems	20%
20.3	Project-type activity	40%
20.4	Open-ended laboratory work	20%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Cloud Computing Technology Fundamentals
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 733
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL331

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	NA

10.	**Frequency of offering**	Alternate year

11.	**Faculty who will teach the course:** Huzur Saran, Sorav Bansal, SC Gupta	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The course aims to cover Fundamentals of Virtualisationand Cloud Technology, including Tools for building cloud and cloud based systems and related disruptive technologies like software defined networks (SDN) and Software defined storage (SDD)

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Overview of Cloud Computing, Virtualisation of CPU, Memory and I/O Devices; Storage Virtualisation and Software Defined Storage (SDS), Software Defined Networks (SDN) and Network Virtualisation, Data Centre Design and interconnection Networks, Cloud Architectures, Public Cloud Platforms (Google App Engine, AWS,Azure), Cloud Security and Trust Management, Open Source Clouds (Baadal, Open Stack, Cloud Stack), Cloud Programming and Software Environments (Hadoop, GFS, Map Reduce, NoSQL systems, Big Table, HBase, Libvirt, OpenVswitch), Amazon (Iaas), Azure(PaaS), GAE (PaaS)

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Overview of Cloud Computing	2
2	Virtualisation of CPU, Memory and I/O Devices	6
3	Storage Virtualisation and Software Defined Storage (SDS)	6
4	Software Defined Networks (SDN) and Network Virtualisation	6
5	Data Centre Design and interconnection Networks	2
6	Cloud Architectures&Plateforms (AWS, Azure, GAE)	3
7	Open Source Clouds (Baadal, Open Stack, Cloud Stack...)	3
8	Cloud Programming and Software Environments (Hadoop, GFS, Map Reduce, NoSQL systems, Big Table, HBase, Libvirt, OpenVswitch)	8
9	Cloud Security and Trust Management	2
10	Project Discussion (as and when required)	4
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities - NA

17. Brief description of laboratory activities

(Students will work in group of 3-4 and do the following lab assignments/projects in Cloud Lab)

Moduleno.	Experiment description	No. of hours
1	Installing & experimenting with Open Stack, Baadal, Cloud Sim	6
2	Installing & experimenting with Hadoop& Map-Reduce	6
3	Designing & Implementing a Module in Baadal	14
	COURSE TOTAL *(14 times 'P')*	28

18. Suggested texts and reference materials

1. Kai Hwang, Geoffrey C Fox, Jack J Dongarra, *“Distributed and Cloud Computing”*,Morgan Kaufmann Publishers
2. Articles / Papers related to various technologies covered in the course

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	Virtual Machines in Baadal Cloud
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course*(Percent of student time with examples, if possible)*

20.1	Design-type problems	20%
20.2	Open-ended problems	
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Software Engineering
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL740
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5
7.	**Pre-requisites** (course no./title)	COL106, COL226

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	MAL745 (old)
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☒ 2nd sem ☐ Alternate year

11.	Faculty who will teach the course 1. SC GUPTA	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To learn Software Engineering concepts and techniques relevant to design and development of large complex Software Systems
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Introduction to Software Engineering, Software Life Cycle models and Processes, Requirement Engineering, System Models, Architectural Design, Abstraction & Modularity, Structured Programming, Object-oriented techniques, Design Patterns, Service Oriented Architecture, User Interface

	Design, Verification and Validation, Reliability, Software Evolution, Project Management & Risk Analysis, Software Quality Management, Configuration Management,Software Metrics, Cost Analysis and Estimation, Manpower Management, Organization and Management of large Software Projects

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Overview of Software Engineering	4
2	Requirement Engineering	6
3	Software Architectural & Detailed Design	6
4	Validation and Verification	4
5	Software Evolution	2
6	Project Management	2
7	Managing People	2
8	Software Metrics, Cost Analysis & Estimation	4
9	Quality Management & Standards	2
10	Configuration Management	2
11	Service Oriented Architecture	2
12	Project Discussion & Review (as and when required)	6
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Project Requirement Specification& Documentation	8
2	Project System Design& documentation	6
3	Project Programming and Testing	14
4		
5		
COURSE TOTAL *(14 times 'P')*		28

18. Suggested texts and reference materials

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1. Software Engineering, Ian Somerville, Pearson Education
2. IEEE Software Engineering related Standards

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

(Students will choose one of the over 10 offered projects in groups of 3-4 and do Requirement, Design, Development & Testing using the software engineering principles taught in the class)

20.1	Design-type problems	20%
20.2	Open-ended problems	
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Foundations of Automatic Verification
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL750
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL226, COL352

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Alternate year

11.	**Faculty who will teach the course** 1. S. Arun-Kumar 2. Sanjiva Prasad

12.	**Will the course require any visiting faculty?**	NO

13.	**Course objective** *(about 50 words)*: Computer-Aided Verification has emerged as an important area in the design of reliable and safety-critical hardware and software systems. Recent advances in theory, practice and technology has made automatic verification tools fast, efficient and widely employed components in the hardware and software development cycles, model-checking being the most common of these techniques. This course aims at providing students with the foundations

	of verification technology as well as extensive hands-on

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: A selection from the following topics, and experiments with the mentioned tools: Review of first-order logic, syntax and semantics. Resolution theorem proving. Binary Decision Diagrams (BDDs) and their use in representing systems. (Programming exercises coding and using logic programming frameworks). Transition systems, automata and transducers. Buechi and other automata on infinite words; Linear Time Temporal Logic (LTL), and specifying properties of systems in LTL; the relationship between temporal logic and automata on infinite words, LTL Model checking (exercises using Spin or similar tools); Computational Tree Logic (CTL and CTL*); CTL model checking (exercises); Process calculi such as CSP and CCS. Notions of program equivalence -- traces, bisimulation and other notions. Hennessy-Milner Logic (HML) and Mu calculus (exercises using tools such as CWB -- Concurrency Work Bench). Symbolic model checking, exercises using tools such as SMV. Sat-based model checking and Davis-Putnam procedure; *(exercises using tools such as nuSMV)*. Possible additional topics include: equational logic frameworks, real-time frameworks, reactive frameworks, pi-calculus (*exercises using tools such as the Mobility Workbench*), Tree automata and Weak Second-order Logic with k successors (WS*k*S), (*exercises using Mona or similar tools)*.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Resolution, BDDs and Model-checking	4
2	Transition Systems, Automata and Transducers	4
3	Buechi Automata and Infinite Words	6
4	Linear Temporal Logic (LTL)	6
5	Relationship between LTL and automata on infinite words	6
6	Computational Tree Logic	6
7	Process Calculi and Hennessy-Milner Logic	6
8	Mu-calculus and other advanced topics	4
9		
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	exercises using Spin or similar tools	5
2	exercises using tools such as CWB -- Concurrency Work Bench	5
3	exercises using tools such as SMV	5
4	exercises using tools such as nuSMV	5
5	Symbolic Model-checking	8
COURSE TOTAL *(14 times 'P')*		28

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. C. A. R. Hoare, Communicating Sequential Processes, Prentice-Hall, 1985.
2. R. Milner, Communication and Concurrency, Prentice Hall, 1989.
3. M. Hennessy, Algebraic Theory of Processes, MIT Press, 1988
4. E. Clarke, O. Grumberg, D. Peled, Model Checking, MIT Press, 2000
5. KL McMillan, Symbolic Model Checking, Kluwer Academic Publishers, 1993.
6. G. Bruns, Distributed Systems Analysis with CCS, Prentice Hall, 1997
7. W Fokkink, Introduction to Process Algebra, Springer, 2000
8. C. Stirling, Modal and Temporal Properties of Processes, Springer, 2001.
9. H. R. Anderson, An Introduction to Binary Decision Diagrams, 1997.
10. J-P Katoen, Concepts, Algorithms and Tools for Model Checking, 1998.
11. G. Holzmann, Design and Validation of Computer Protocols, Prentice-Hall, 1991

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	30%
20.2	Open-ended problems	10%
20.3	Project-type activity	30%
20.4	Open-ended laboratory work	10%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Algorithmic Graph Theory
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	CSL 751
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 351 (Analysis and Design of Algorithms)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	60% overlap with MAL656 (old Maths course)
8.3	Supersedes any existing course	-

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every sem

11.	**Faculty who will teach the course**	Naveen Garg, Amit Kumar
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: This course is meant to be a graduate level introduction to algorithmic techniques for graph theoretic problems. The emphasis would be on designing efficient algorithms and to identify graph parameters/classes for which even hard problems can be solved efficiently.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Algorithms for computing maximum s-t flows in graphs: augmenting path, blocking flow, push-relabel, capacity scaling etc. Max-flow min-cut theorem and its applications Algorithms for computing min-cuts in graphs, structure of min-cuts. Min-cost flows and applications: cycle cancelling algorithms, successive shortest paths, strongly polynomial algorithms. Maximum matching in bipartite and general graphs: Hall's theorem, Tutte's theorem, Gallai-Edmonds decomposition. Weighted bipartite matching, Edmonds Algorithm for Weighted Non-Bipartite Matching,T-joins and applications. Factor-Critical Graphs, Ear Decompositions, Graph orientations, Splitting Off, k-Connectivity Orientations and Augmentations, Arborescences and Branchings, Edmonds therorem for disjoint arborescences. Planar graphs, algorithms for checking planarity, planar-separator theorem and its applications. Intersection graphs, perfect graphs: polyhedral characterization, the strong perfect graph theorem, kinds of perfect graphs and algorithms on them. Treewidth, algorithm for computing tree width, algorithms on graphs with bounded tree width.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Algorithms for computing maximum s-t flows in graphs: augmenting path, blocking flow, push-relabel, capacity scaling etc.	7
2	Max-flow min-cut theorem and its applications	3
3	Algorithms for computing min-cuts in graphs, structure of min-cuts	3
4	Min-cost flows and applications: cycle cancelling algorithms, successive shortest paths, strongly polynomial algorithms	4
5	Maximum matching in bipartite and general graphs: Hall's theorem, Tutte's theorem, Gallai-Edmonds decomposition.	3
6	Weighted bipartite matching, Edmonds Algorithm for Weighted Non-Bipartite Matching,T-joins and applications	4
7	Factor-Critical Graphs, Ear Decompositions, Graph orientations, Splitting Off,	5

	k-Connectivity Orientations and Augmentations, Arborescences and Branchings, Edmonds therorem for disjoint arborescences	
8	Planar graphs, algorithms for checking planarity, planar-separator theorem and its applications.	3
9	Intersection graphs, perfect graphs: polyhedral characterization, the strong perfect graph theorem, kinds of perfect graphs and algorithms on them.	6
10	Treewidth, algorithm for computing tree width, algorithms on graphs with bounded tree width.	4
	COURSE TOTAL (14 times 'L')	**42**

16.	**Brief description of tutorial activities: NA**

17.	**Brief description of laboratory activities: NA**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Reinhard Diestel. Graph Theory. Springer-Verlag 2010
2. T. Kloks. Advanced Graph Algorithms.
3. Alexander Schrijver. A course in combinatorial Optimization. Springer 2012

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	NA
19.2	Hardware	NA
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Geometric Algorithms
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 752
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 351 (Algorithm Design and Analysis) or equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	

9.	**Not allowed for** (indicate program names)	

10.	Frequency of offering	Either Sem

11.	**Faculty who will teach the course**	Sandeep Sen, Amit Kumar, Ragesh Jaiswal, Amitabha Bagchi
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: The course is directed at designing algorithms for problems that are geometric that have applications including CAD/CAM, robotics, computer graphics, molecular biology, GIS, spatial databases, sensor networks, and machine learning. The course will emphasize several specialized data representations and data structures for such problems as well as relevant combinatorial geometric techniques that helps us in analysing the behavior of such algorithms. .

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Geometric Fundamentals: Models of computation, lower bound techniques, geometric primitives, geometric transforms Convex hulls: Planar convex hulls, higher dimensional convex hulls, randomized, output-sensitive, and dynamic algorithms, applications of convex hull, Intersection detection: segment intersection, line sweep, map overlay, halfspace intersection, polyhedra intersection, Geometric searching: segment, interval, and priority-search trees, point location, persistent data structure, fractional cascading,

	range searching, nearest-neighbor searching Proximity problems: closest pair, Voronoi diagram, Delaunay triangulation and their subgraphs, spanners, well separated pair decomposition Arrangements: Arrangements of lines and hyperplanes, sweep-line and incremental algorithms, lower envelopes, levels, and zones, applications of arrangementsTriangulations: monotone and simple polygon triangulations, point-set triangulations, optimization criteria, Steiner triangulation, Delaunay refinementGeometric sampling: random sampling and ε-nets, ε-approximation and discrepancy, cuttings, coresetsGeometric optimization: linear programming, LP-type problems, parametric searching, approximation techniques. Implementation Issues : robust computing, perturbation techniques, floating-point filters, rounding techniques

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction and Visibility Problems	2
2	Plane sweep technique and applications	3
3	Convex hulls, Quickhull, Dynamic maintenance, Output sensitive algorithms	5
4	Dual Transformation, Intersection of half-planes	2
5	Lower bounds in Algebraic decision trees	2
6	Triangulation and Planar Point Location, Persistent Data Structures	5
7	Voronoi Diagrams, Delaunay Triangulation, Lifting Transform and Convex hulls	5
8	Randomized Incremental construction and random Sampling	4
9	Arrangements and Levels	3
10	Range Searching : Orthogonal and Simplex, Range Trees, Partition Trees	4
11	Epsilon nets and approximations, Quadtrees and applications to Well-separated Pair decomposition, Geometric Optimization including fixed dimensional linear programming, Approximation Algorithms for ETSP, Set Cover	7
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities -** NA

17.	Brief description of laboratory activities: Implementation of Convex hulls, Delaunay Triangulation, Range Trees, Robust implementation of geometric algorithms because of finite precision arithmetic, a short Project in groups of 3-4 students of a real life problem

Module no.	Experiment description	No. of hours
1	Planar convex hull	6
2	Voronoi diagrams/Delaunay Triangulation	6
3	K-d trees/Quadtrees	6
4	A mini project (in groups)	10
COURSE TOTAL ***(14 times 'P')***		**28**

18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. **Computational Geometry** by M. de Berg, M. van Kreveld, M. Overmars, and O. Schwarzkopf, pub: Springer
2. Computational Geometry in C, J. O' Rourke, Cambridge University Press
3. Algorithms in Combinatorial Geometry, H. Edelsbrunner, pub: Springer-Verlag (EATCS Monograph)

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download. Matlab for doing assignments.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: The course is designed to be a mix of theory as well as hands on experience. A background in Linear Algebra and Basic Probability is required.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Complexity Theory
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 753
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL352 (Theory of Computation)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Once every four semesters

11.	**Faculty who will teach the course**	Ragesh Jaiswal, Sandeep Sen, Amitabha Bagchi, Naveen Garg, Amit Kumar
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: This course is meant to be an introductory course on Computational Complexity Theory aimed at advanced undergraduate and graduate students.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Modeling computation (Finite state machines, Non-determinism, Turing machines, class P etc.), NP and NP-completeness, Diagonalization (Time hierarchy and Ladner's theorem), Space complexity (PSPACE, NL, Savitch's theorem, Immerman-Szelepcsényi theorem etc.), Polynomial hierarchy, Boolean circuits (P/poly), Randomized classes (RP, BPP, ZPP, Adleman's Theorem, Gács-Sipser-Lautemann Theorem), Interactive proofs (Arthur-Merlin, IP=PSPACE), Cryptography (one-way functions, pseudorandom generators, zero knowledge), PCP theorem and hardness of approximation, Circuit lower bounds (Hastad's switching lemma), Other topics (#P, Toda's theorem, Average-case complexity, derandomization, pseudorandom construction).

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Modeling Computation	3
2	NP and NP-Completeness	3
3	Space Complexity	3
4	Polynomial Hierarchy	4
5	Boolean circuits	3
6	Randomized classes	4
7	Interactive proofs	4
8	Cryptography	3
9	PCP Theorem and Hardness of approximation	6
10	Other topics (counting classes, Average-case complexity, derandomization, pseudorandom constructions)	9
11		
COURSE TOTAL	***(14 times 'L")***	**42**

16.	**Brief description of tutorial activities:** NA

17.	**Brief description of laboratory activities:** NA

18.	**Suggested texts and reference materials**

1. Sanjeev Arora and Boaz Barak, Computational Complexity: A Modern Approach, Cambridge Universtiy Press, 2009.
2. Oded Goldreich, Computational Complexity: A Conceptual Prespective, Cambridge University Press, 2008.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	100% This is a theoretical course. The students will be working mainly on problems that help in understanding the concepts.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering	
2.	**Course Title** *(< 45 characters)*	Approximation algorithms	
3.	**L-T-P structure**	3-0-0	
4.	**Credits**	3	
5.	**Course number**	COL 754	
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5	

7.	**Pre Requisites** (course no./title)	COL 351

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	-

11.	**Faculty who will teach the course**	Naveen Garg, Amit Kumar
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)***:** This course is intended to give students tools for designing algorithms for computationally hard problems. These algorithms will run in polynomial time and will

	come close to the optimal solution. The students will learn sophisticated tools from linear and convex programming for designing and analyzing such algorithms.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: NP-hardness and approximation algorithms. Different kinds of approximability. Greedy algorithm and local search with applications in facility location, TSP and scheduling. Dynamic programming with applications in knapsack, Euclidean TSP, bin packing. Linear programming, duality and rounding. Applications in facility location, Steiner tree and bin packing. Randomized rounding with applications. Primal-dual algorithms and applications in facility location and network design. Cuts and metrics with applications to multi-commodity flow. Semi-definite programming and applications: max-cut, graph coloring. Hardness of approximation.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Approximation algorithms – notion and basic examples	2
2	Greedy algorithms: scheduling, TSP, Steiner Tree	4
3	Dynamic programming: knapsack, bin packing, Euclidean TSP, makespan on parallel machines	6
4	Linear programming and duality	2
5	Basic examples of rounding: facility location, vertex cover	2
6	Randomized Rounding: max-sat, chernoff bounds, multi-commodity flow	4
7	Primal-dual algorithms: network design, facility location, k-median	6
8	Multicut, probabilistic approximation by tree metrics and applications	4
9	Multi-commodity flow problems and metric embeddings	4
10	Semidefinite programming: max flow, graph coloring	4
11	Hardness of approximation	4
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities:** NA

17.	**Brief description of laboratory activities:** NA

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. David P. Williamson and David B. Shmoys, The Design and Analysis of Approximation Algorithms, 1st edition, Cambridge University Press, 2011
2. Vijay V. Vazirani, Approximation Algorithms, 2nd edition, Springer, 2002.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	10%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Mathematical Programming
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 756
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	**COL351 (Analysis and Design of Algorithms)**

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	40% overlap with MAL210 (Optimization methods and applications), MAL638 (Applied non-linear programming) and MAL726 (Principles of Optimization theory). 60% overlap with Numerical Optimization
9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	-

11.	**Faculty who will teach the course**	Naveen Garg, Amit Kumar
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be a graduate level introduction to linear, non-linear and convex optimization and integer programming. The emphasis will be on modeling optimization problems and on efficient algorithms for solving them.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Linear programming: introduction, geometry, duality, sensitivity analysis. Simplex method, Large scale optimization, network simplex. Ellipsoid method, problems with exponentially many constraints, equivalence of optimization and separation. Convex sets and functions – cones, hyperplanes, norm balls, generalized inequalities and convexity, perspective and conjugate functions. Convex optimization problems – quasi-convex, linear, quadratic, geometric, vector, semi-definite. Duality – Lagrange, geometric interpretation, optimality conditions, sensitivity analysis. Applications to approximation, fitting, statistical estimation, classification. Unconstrained minimization, equality constrained minimization and interior point methods. Integer Programming: formulations, complexity, duality. Lattices, geometry, cutting plane and branch and bound methods. Mixed integer optimization

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Linear programming: introduction, geometry, duality, sensitivity analysis	6
2	Simplex method, Large scale optimization, network simplex	4
3	Ellipsoid method, problems with exponentially many constraints, equivalence of optimization and separation	4
4	Convex sets and functions – cones, hyperplanes, norm balls, generalized inequalities and convexity, perspective and conjugate functions,	4
5	Convex optimization problems – quasi-convex, linear, quadratic, geometric, vector, semi-definite. Duality – Lagrange, geometric interpretation, optimality conditions, sensitivity analysis	4
6	Applications to approximation, fitting, statistical estimation, classification	5
7	Numerical linear algebra background, unconstrained minimization, equality constrained minimization and interior point methods	7
8	Integer Programming: formulations, complexity, duality	3
9	Lattices, geometry, cutting plane and branch and bound methods	3

10	Mixed integer optimization	2
	COURSE TOTAL (14 times 'L')	**42**

16.	**Brief description of tutorial activities: NA**

17.	**Brief description of laboratory activities: NA**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Bertsimas, Dimitris, and Tsitsiklis, John. Introduction to Linear Optimization. Belmont, MA: Athena Scientific, 1997. ISBN: 9781886529199.
2. Boyd, Stephen and Vandenberghe, Lieven. Convex Optimization. Cambridge University Press.
3. Bertsimas, Dimitris, and Weismantel, Robert. Optimization over Integers. Dynamic Ideas 2005.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Matlab for doing assignments.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%

20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Model Centric Algorithm Design
3	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 757
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 351 (Algorithm Design and Analysis) or equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	-

11.	**Faculty who will teach the course**	Sandeep Sen, Amit Kumar, Naveen Garg, SN Maheshwari
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)***:** The aim of the course is to make students familiar with contemporary algorithmic paradigms that are used to solve real life problems in frameworks not compatible with the Random Access Machine model, for example multicore models, streaming data, and caching issues. Since the design metrics and different, many new issues have to be addressed to design good algorithms in such computing environments.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)***:** The RAM model and its limitations, Introduction to alternate algorithmic models Parallel models like PRAM and Interconnection networks; Basic problems like Sorting, Merging, Routing, Parallel Prefix and applications, graph algorithms like BFS, Matching, Memory hierarchy models: Caching, Sorting, Merging, FFT, Permutation, Lower bounds Data

	Structures - searching, Priority queues Streaming Data models: Distinct items, frequent items, frequency moments, estimating norms, clustering, On line algorithms: competitive ratio, list accessing, paging, k-server, load-balancing, lower-bounds

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Idealized RAM model and its limitations	2
2	On line problems like list access, Competitive ratio	2
3	Paging :Deterministic strategy, lower bound, Randomized strategy	4
4	Load balancing, k-server problem	4
5	Parallel Computing models : shared memory, Interconnection networks	2
6	Sorting and selection in parallel models, Prefix Computation and applications	9
7	Matrix problems, BFS in graphs, Connectivity	4
8	Streaming Data model : read once, sublinear space sketch, Approximate expected bounds	3
9	Distinct elements, Frequent elements, random sampling, Frequency moment problem	6
10	Memory hierarchy models : Matrix Transpose, Sorting, FFT	4
11	Cache oblivious algorithms for searching, matrix transpose	2
COURSE TOTAL ***(14 times 'L')***		**42**

16.	Brief description of tutorial activities - NA
17.	Brief description of laboratory activities: Take up some classic prototype problem in each of the models and actually measure the performance of the specially Designed algorithms. This may be done individually or in small groups.

Module no.	Experiment description	No. of hours
1	Group project on some suitable on-line problem	8
2	Group project on a suitable streaming data problem	8
3	Implementation of sorting in multicore machine using openMP or MPI	6
4	Comparing cache misses between a cache sensitive and RAM based algorithm for matrix transpose (using processor specific tools to measure L1, L2 cache misses	6
COURSE TOTAL ***(14 times 'P')***		**28**

18.	Suggested texts and reference materials STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Joseph Ja Ja, An Introduction to Parallel Algorithms, Addison Wesley1992.
2. Borodin and El-Yaniv, Online Computation and Competitive Analysis, Cambridge University Press, 1998.
3. J.S Vitter, Lecture notes on External Memory Algorithms and Data Structures, Survey article in ACM Computing Survey, Volume 33 Issue 2, June 2001 Pages 209-271
4. Muthukrushnan, Data Streams: Algorithms and Applications, Foundations and Trends® in Theoretical Computer Science

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	LEDA platform including the EM extension
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: The course is designed to be a mix of theory as well as hands on experience. A background in Basic Probability is required.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Algorithms
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 758
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL 351 or equivalent

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	No
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	☐Every sem ☐1st sem ☐2nd sem ☒Alternate year

11.	**Faculty who will teach the course** Sandeep Sen, Amit Kumar, Naveen Garg, SN Maheshwari, Ragesh Jaiswal	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Advanced topics in various areas of algorithm design laying the foundations to pursue research in this area

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Advanced data structures: self-adjustment, persistence and multidimensional trees. Randomized algorithms: Use of probabilistic inequalities in analysis, Geometric algorithms: Point location, Convex hulls and Voronoi diagrams, Arrangements applications using examples. Graph algorithms: Matching and Flows. Approximation algorithms: Use of Linear programming and primal dual, Local search heuristics. Parallel algorithms: Basic techniques for sorting, seraching, merging, list ranking in PRAMs and Interconnection networks.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Review of Algorithmic Techniques	3
2	Graph Algorithms	7
3	Randomized Algorithms	8
4	Parallel and Distributed Algorithms	7
5	NP-completeness and Approximation Algorithms	8
6	Online algorithms	5
7	Geometric Algorithms	4
8		
9		
10		
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities - NA

17. Brief description of laboratory activities

Moduleno.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
	COURSE TOTAL *(14 times 'P')*	

18. Suggested texts and reference materials

1. Motwani R, Raghavan P., Randomized Algorithms, Cambridge University Press
2. Preparata, Shamos, Computational Geometry, Springer Verlag.
3. Mehlhorn K., Data Structures and Algorithms:1Searching and Sorting, Springer Verlag EATCS Monograph on Theoretical Computer Science
4. Papadimitrou, Steiglitz, Combinatorial Optimization, Princeton University Press
5. Ja'Ja' J., Introduction to Parallel Algorithms, Addison-Wesley
6. Vazirani, Approximation Algorithms, Springer Verlag

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Cryptography and Computer Security
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 759
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL351 or equivalent and MAL250 or equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	MAL724 (30%) MAL730 (50%) MAL786 (50%)
8.3	Supersedes any existing course	No

9.	**Not allowed for** (indicate program names)	N.A.

10.	**Frequency of offering**	Every Alternate semester

11.	**Faculty who will teach the course**	Ragesh Jaiswal, Shweta Agrawal,
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The course offers an introduction to modern methods of cryptography. Cryptography is a rigorous study of the science of secret keeping. The course initiates students in distilling real world security problems into mathematical models and reasoning about the security of real world systems and protocols using these models. Rigorous guarantees of security require that breaking a cryptosystem be provably as hard as solving some well known difficult mathematical problem, such as factoring. The course studies these methods via various practically motivated examples such as encryption systems, digital signatures, multiparty computation protocols etc.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Part 1: Foundations: Perfect secrecy and its limitations, computational security, pseudorandom generators and one time encryption, pseudorandom functions, one way permutations, message authentication and cryptographic hash functions. Part 2: Basic Constructions and proofs: Some number theory, symmetric key encryption, public key encryption, CPA and CCA security, digital signatures, oblivious transfer, secure multiparty computation. Part 3: Advanced Topics: Zero knowledge proofs, identity based encryption, broadcast encryption, homomorphic encryption, lattice based cryptography.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction – relevance of cryptography to real world security, principles of cryptography: defining security, concrete mathematical assumptions, reductions to assumptions. Case study of secure encryption : how to define security, understanding what assumptions are needed.	2
2	Perfect and Computational indistinguishability, pseudorandom generator, hybrid arguments, one way permutations, hardcore predicates, one way functions, Goldreich Levin predicate.	4
3	Stream Ciphers, Block Ciphers, Pseudorandom functions, PRF from PRG	3
4	Hash functions: universal, collision resistant, universal one way, MACs	3

5	Public key Encryption: Definitions: IND and SIM based security, CPA and CCA security. Constructions: RSA, DDH, QR, Lattice CPA secure PKE from Trapdoor OWP Cramer Shoup CCA secure cryptosystem	8
6	Identity Based Encryption	3
7	Digital Signatures from OWF, from DDH, Lattices	3
8	Zero Knowledge: interactive/non-interactive, statistical/computational -- definitions and constructions	4
9	Protocols: oblivious transfer, bit commitment, coin flipping, secret sharing, BitCoins	4
10	Secure Multiparty computation : Yao's garbled circuits	3
11	Homomorphic encryption, functional encryption, broadcast encryption.	5
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities -** NA
17.	**Brief description of laboratory activities -** NA

Module no.	**Experiment description**	**No. of hours**
COURSE TOTAL ***(14 times 'P')***		

18.	**Suggested texts and reference materials** 1. Introduction to Modern Cryptography by Jonathan Katz and Yehuda Lindell. Chapman & Hall/CRC, 2007. 2. Lecture notes on Cryptography by Shafi Goldwasser and Mihir Bellare. Available on author's website: http://cseweb.ucsd.edu/users/mihir/papers/gb.html

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	N.A.
19.2	Hardware	Personal laptop or Computer access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	N.A.
20.2	Open-ended problems	N.A.
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	N.A.
20.5	Others (please specify)	80%. This is primarily a theoretical course. The students will be working mainly on problems that help in understanding the concepts.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Data Management
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 760
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 362 (Introduction to Database Systems)

8.	**Status vis-à-vis other courses** *(give course number/title)*	NA
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every sem / Alternate sem

11.	**Faculty who will teach the course**	Maya Ramanath, S K Gupta
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The course will introduce advanced data management techniques with the main focus being on the fundamentals of distributed data management, and the management of massive amounts of data (or "Big Data").

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Storage and file structures, advanced query processing and optimization for single server databases, distributed data management (including distributed data storage, query processing and transaction management), web-data management (including managing the web-graph and implementation of web-search), big data systems.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Database Architectures (client-server, distributed/parallel, cloud)	3
2	Storage and file structures	4
3	Advanced query processing and optimization	4
4	Distributed Data Storage	2
5	Distributed Query Processing and Optimization	8
6	Distributed Transaction Management	4
7	Web Data Management (Managing the Web Graph, Web Search)	6
8	Big Data Systems and data processing (Introduction and Motivation, Mapreduce, Pregel, Hive, Pig Latin, etc.)	11
9		
10		
11		
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities** - NA

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Query Optimization for Complex Queries	4
2	Learning about the fundamentals of distributed query processing	4
3	Implementation of Web-search algorithms	10
4	Programming with big-data systems	10
COURSE TOTAL *(14 times 'P')*		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, "Database Systems: The Complete Book", 2nd edition, Prentice Hall, 2008.
2. Tamer Oszu, Patrick Valduriez, "Principles of Distributed Database Systems", 3rd edition, Springer, 2010.
3. Journal and conference papers.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Java. Perl/PHP. PostgreSQL. Webserver, Hadoop/Giraph installations, Galago/Lucene installation
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector

19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Database Implementation
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 762
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL 362 (Introduction to Database Systems)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	Nil
8.2	Overlap with any UG/PG course of other Dept./Centre	Nil

9.	**Not allowed for** (indicate program names)	-

10.	**Frequency of offering**	Every sem **/** Either sem

11.	**Faculty who will teach the course**	S. K. Gupta, Maya Ramanath
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The course addresses the issue of database implementation, query processing and query optimization for relational databases. Different file structures, indexing and hashing techniques and query processing will be addressed.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Review of Relational Model, Algebra and SQL, File structures, Constraints and Triggers, System Aspects of SQL, Data Storage, Representing Data Elements, Index, Multi dimensional and Bit-map Indexes, Hashing, Query Execution, Query Compiler.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topics	No. of hours
1	Review of Relational Model, Algebra and SQL	3
2	Constraints and Triggers	3
3	File Structures	6
4	Data Storage	4
5	Representing Data Elements	4
6	Index, Multi dimensional and Bit-map Indexes	10
7	Query Execution	6
8	Query Compiler	6
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities** - NA

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	SQL Query Parsing and Compilation	6
2	Transformation of SQL into relational algebra	6
3	Different Algorithm for Relational Algebra Operation	8
4	Query Processing and Optimization	8
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Hector Garcia-Molina, Jeffrey D. Ullman, Jennifer Widom, "Database Systems: The Complete Book", 2nd edition, Prentice Hall, 2008
2. Tamer Oszu, Patrick Valduriez, "Principles of Distributed Database Systems", 3rd edition, Springer, 2010
3. Journal and conference papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Java. Perl/PHP. PostgreSQL. Webserver, mySQL
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Department of Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Introduction To Logic and Functional Programming
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL765
6.	**Status** (*category* for *program*)	PC for MCS, JCA

7.	**Pre-requisites** (course no./title)	Nil.

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	25% with COL703, minor overlap with COL226
8.2	Overlap with any UG/PG course of another Dept./Centre	

9.	**Not allowed for** (indicate program names)	Not allowed for B. Techs and Dual Degree students of any programme.

10.	**Frequency of offering**	Every odd semester

11.	**Faculty who will teach the course:** Saroj, Sanjiva, S Arun Kumar	
12.	**Will the course require any visiting faculty?**	No
13.	**Course objective** *(about 50 words; motivation and aims of this course)***:** The objective of the course is to introduce applicative/declarative style of computing based on mathematical principles. There are mainly two computing paradigms in the applicative/declarative category, one based on resolution and the other on reduction. Logic and Functional languages are associated with these two important classes of declarative languages.	

14.	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)*: Introduction to declarative programming paradigms. The functional style of programming, paradigms of developments of functional programs, use of higher order functionals and pattern-matching. Introduction to lambda calculus. Interpreters for functional languages and abstract machines for lazy and eager lambda calculi, Types, type-checking and their relationship to logic. Logic as a system for declarative programming. The use of pattern-matching and programming of higher order functions within a logic programming framework. Introduction to symbolic processing. The use of resolution and theorem-proving techniques in logic programming. The relationship between logic programming and functional programming.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Introduction to declarative paradigms	1
2	Functional programming: data types and higher-order functions	8
3	Type-systems and type inference	3
4	Lambda calculus and implementation	6
5	Eager and lazy evaluation	4
6	Review of Propositional logic, syntax and semantics	3
7	Predicate Logic, syntax and semantics	5
8	Resolution and other proof techniques	4
9	Prolog and relational programming	6
10	Theorem Proving basics	2
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16.	**Brief description of tutorial activities**

17. Brief description of laboratory activities

Module no.	**Experiment description**	**No. of hours**
1	Programming with inductive data types: lists, trees and abstract types	4
2	Higher-order functions	4
3	Lambda calculus interpreters	4
4	Terms and unification	4

5	Type checking	4
6	Prolog programming	4
7	Prolog interpreter	4
8		
9		
10		
	COURSE TOTAL *(14 times 'P')*	28

18. Suggested texts and reference materials

1. Laurence C. Paulson. ML for the Working Programmer. Cambridge University Press.
2. Chris Reade. Elements of Functional Programming. Addison-Wesley.
3. Peter Henderson. Functional Programming: Applications and Implementation. Prentice Hall.
4. J. W. Lloyd. Foundations of Logic programming. Springer-Verlag. New York.
5. John Kelly, The Essence of Logic, Prentice Hall of India, 1997.
6. Anil Nerode and Richard A. Shore. Logic for Applications. Springer-Verlag.
7. Leon Sterling and Ehud Shapiro. The Art of Prolog (Advanced Programming Techniques), Prentice Hall of India, 1996.
8. Saroj Kaushik, Logic and Prolog Programming, New Age International 2002.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	SML or OCaml interpreter, Prolog interpreter
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	
20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	Design of declarative language
20.2	Open-ended problems	Integrating logic and functional programming styles
20.3	Project-type activity	Implementing functional and logic interpreters
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Department of Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Wireless Networks
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL768
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL334 or equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	N/A
8.2	Overlap with any UG/PG course of another Dept./Centre	N/A

9.	**Not allowed for** (indicate program names)	N/A

10.	**Frequency of offering**	Every alternative semester

11.	**Faculty who will teach the course** 1. V. Ribeiro, 2. Huzur Saran, 3. A. Seth	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words; motivation and aims of this course)***:** Wireless technology is rapidly improving and has a large potential for transforming our nation. This course will help students keep abreast of the state-of-the-art wireless networking technology at all protocol layers.

14.	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)*: Radio signal propagation, advanced modulation and coding, medium access techniques, self-configurable networks, mesh networks, cognitive radio and dynamic spectrum access networks, TCP over wireless, wireless security, emerging applications.

15.	**Lecture Outline** *(with topics and number of lectures)*	
Module no.	**Topic**	**No. of hours**
1	Radio signal propagation	4
2	Advanced modulation and coding	6
3	Medium access techniques	8
4	Routing in multihop wireless networks	8
5	TCP over wireless	4
6	Wireless security	6
7	Emerging applications	6
8		
9		
10		
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16.	**Brief description of tutorial activities** - NA	
17.	**Brief description of laboratory activities**	
Module no.	**Experiment description**	**No. of hours**
1	Course project: software defined radio, network simulations, sensor network deployment etc.	28
2		
3		
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6		
7		
8		
	COURSE TOTAL *(14 times 'P')*	28

18.	Suggested texts and reference materials 1. Wireless Communications - Andreas F. Molisch, John Wiley and Sons, 2005 (Indian Edition) 2. Wireless Networking – J. L. Burbank et al., John Wiley and Sons, 2013

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	Open network simulators (e.g. ns-2, ns-3), proprietary simulators (e.g. Qualnet)
19.2	Hardware	Software defined radios (e.g. USRP), sensor motes
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	Spectrum analyzers
19.6	Classroom infrastructure	Whiteboard, LCD projector
19.7	Site visits	
20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	30%
20.2	Open-ended problems	
20.3	Project-type activity	70%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Artificial Intelligence
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 770
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL 106 (Data Structures)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	50% Overlap with CSL 333/CSL 671 (Fundamentals of Artificial Intelligence)
8.2	Overlap with any UG/PG course of other Dept./Centre	Partial Overlap with EE 758.
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every sem

11.	**Faculty who will teach the course**	Mausam, Saroj Kaushik, Parag Singla, KK Biswas
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course introduces several advanced ideas in the field of artificial intelligence. It is tailored to students who have already had some exposure of the subject, either as an undergraduate AI course or an online course. Since AI is quite a broad field, this course will take an overview approach, both of traditional and modern AI. Since the students would have had basic exposure of AI, the course will move fast and cover a lot more material than CSL 333.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*:
	Philosophy of artificial intelligence, fundamental and advanced search techniques (A*, local search, suboptimal heuristic search, search in AND/OR graphs), constraint optimization, temporal reasoning, knowledge representation and reasoning through propositional and first order logic, modern game playing (UCT), planning under uncertainty (Topological value iteration, LAO*, LRTDP), reinforcement learning, introduction to robotics, introduction to probabilistic graphical models (Bayesian networks, Hidden Markov models, Conditional random fields), machine learning, introduction to information systems (information retrieval, information extraction).

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Course Outline; Philosophy of Machine Intelligence	3
2	Uninformed, Heuristic and Local Search	3
3	Search in AND/OR Graphs	2
4	Constraint Optimization	3
5	Temporal Reasoning	2
6	Knowledge Representation: Propositional Logic	3
7	First order logic and Logic programming	5
8	Game Playing and UCT algorithm	3
9	Planning under Uncertainty	3
10	Reinforcement learning	3
11	Probabilistic Graphical models (directed & undirected)	5
12	Machine Learning and Neural Networks	3
13	Introduction to Robotics	2
14	Introduction to Information Systems	2
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities -** NA

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	Modeling problems as AND/OR search	5
2	Modeling problems using Prolog	6
3	Using UCT to solve large games	5
4	Solving problems using planning under uncertainty algorithms	6
5	Modeling problems using conditional random fields	6
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Stuart Russell & Peter Norvig, “Artificial Intelligence: A Modern Approach”,
1. Prentice-Hall, Third Edition (2009)
2. Saroj Kaushik, “Artificial Intelligence”, Cengage Learning, 2011.
3. Tom Mitchell. “Machine Learning” First Edition, McGraw-Hill, 1997.
4. Online Reading Material.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download (e.g. Weka machine learning repository, UCI datasets). Matlab may be used for couple of assignments.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	10%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Natural Language Processing
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 772
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL106 (Data Structures) or Equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every sem / Either sem

11.	**Faculty who will teach the course**	Mausam, Saroj Kaushik, Maya Ramanath
12.	**Will the course require any visiting faculty?**	No

13. Course objective *(about 50 words)*:

This course teaches various basic and advanced ideas in the field of natural language processing. The students will learn fundamental algorithms for processing natural languages such as English. The course will provide all requisite background in machine learning and probabilistic graphical models before describing their applications to NLP. The students will also learn how these techniques can be used to solve various practical problems such as information search, question answering, document summarization, etc.

14. Course contents *(about 100 words) (Include laboratory/design activities)*:

NLP concepts: Tokenization, lemmatization, part of speech tagging, noun phrase chunking, named entity recognition, coreference resolution, parsing, information extraction, sentiment analysis, question answering, text classification, document

	clustering, document summarization, discourse, machine translation. Machine learning concepts: Naïve Bayes, Hidden Markov Models, EM, Conditional Random Fields, MaxEnt Classifiers, Probabilistic Context Free Grammars.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Course Outline	1
2	Background: Naïve Bayes, Perceptron, Linear models	3
3	Text Classification	2
4	Language Models	2
5	Tokenization and Lemmatization	2
6	Background: HMMs and CRFs	4
7	NLP pipeline: POS, NP chunking, NER, Coreference	5
8	Background: PCFGs	2
9	Syntactic and Dependency Parsing	3
10	Information Extraction	3
11	Sentiment Analysis	2
12	Information Retrieval	2
13	Background: Latent Dirichlet Allocation	2
14	Document Clustering	2
15	Machine Translation	2
16	Question Answering	2
17	Document Summarization	2
18	Wrap up	1
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

N.A.

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	Classifying a document into known classes	5
2	Implementing a part of speech tagger	5
3	Implementing a relation extractor	5
4	Project	13
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. D. Jurafsky & James H. Martin, Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition, Prentice Hall, Second Edition, 2009.
2. C.D. Manning & H. Schuetze, Foundations of Statistical Natural Language Processing, Cambridge: MIT Press, 1999.
3. James F. Allen: Natural language understanding (2nd ed.). Benjamin/Cummings 1994.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download (e.g. Weka machine learning repository).
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	10%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Machine Learning
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 774
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	Introduction to Probability Theory and Stochastic Processes

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	60% Overlap with CSL 341 (Fundamentals of Machine Learning)
8.2	Overlap with any UG/PG course of other Dept./Centre	60% Overlap with EE 709 (Pattern Recognition). 40% overlap with MAL 803 (Pattern Recognition).

9.	**Not allowed for** (Indicate program names)	CSL 341 (UG - Machine Learning)

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	K KBiswas, Subhashis Banerjee, ParagSingla, Mausam, ManikVarma
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be the first graduate level course in Machine Learning. The field of Machine learning deals with building predictive models of data based on past experience. The subject has become particularly popular in the last decade or so with a wide range of applications. This course will focus on providing students with an in depth understanding of the field including choice of learning models, a variety of learning algorithms, relative merits of various algorithms and examples of practical applications. The course will also provide the required background for carrying out further research in the area.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Supervised learning algorithms: Linear and Logistic Regression, Gradient Descent, Support Vector Machines, Kernels, Artificial Neural Networks, Decision Trees, ML and MAP Estimates, K-Nearest Neighbor, Naive Bayes, Introduction to Bayesian Networks. Unsupervised learning algorithms: K-Means clustering, Gaussian Mixture Models, Learning with Partially Observable Data (EM). Dimensionality Reduction and Principal Component Analysis. Bias Variance Trade-off. Model Selection and Feature Selection. Regularization. Learning Theory. Introduction to Markov Decision Processes. Application to Information Retrieval, NLP, Biology and Computer Vision. Advanced Topics.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Course Outline; Basics	2
2	Linear Regression, Gradient Descent, Logistic Regression	6
3	Linear Support Vector Machines	5
4	Kernels, Non-Linear SVMs	4
5	Artificial Neural Networks, Decision Trees	4
6	Naive Bayes, Introduction to Bayesian Networks	4
7	K-Means, Gaussian Mixture Models, EM	5
8	Feature Selection, Model Selection, Regularization, Bias and Variance	3
9	Learning Theory	3
10	Markov Decision Processes	3
11	Applications and Advanced Topics	3

COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

N.A.

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	Implementation of Linear Regression, Gradient Descent (Batch and Stochastic), SVMs	6
2	Naive Bayes for Text Classification, Decision Trees, Playing with WEKA datasets	6
3	K-Means, PCA for dimensionality Reduction	6
4	Course Project	10
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Kevin P. Murphy. **"**Machine Learning: A Probabilistic Perspective". MIT Press, 2012.
2. Christopher Bishop. "Pattern Recognition and Machine Learning", First Edition,Springer, 2006.
3. R. Duda, P. Hart and D. Stork. "Pattern Classification" Second Edition, John Wiley & Sons INC., 2001.
4. Tom Mitchell. "Machine Learning" First Edition, McGraw-Hill, 1997.
5. Online Reading Material.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download (e.g. Weka machine learning repository, UCI datasets). Matlab for doing assignments.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the Course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: The course is designed to be a mix of theory as well as hands on experience. A background in Linear Algebra and Basic Probability is required.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Learning Probabilistic Graphical Models
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 776
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	Probability Theory and Stochastic Processes

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	-
8.2	Overlap with any UG/PG course of other Dept./Centre	-
8.3	Supersedes any existing course	-

9.	**Not allowed for** (Indicate program names)	-

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	Parag Singla, Mausam, KK Biswas
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be the first graduate level course in the area of Probabilistic Graphical Models (PGM). PGMs have emerged as a very important research field during last decade or so with wide range of applications including Computer Vision, Information Retrieval, Natural Language Processing, Biology and Robotics. This course aims to provide students with a comprehensive overview of PGMs. The course content will include introduction to Probabilistic Graphical Models, directed and undirected representations, inference and learning algorithms and practical applications. This course is also meant to provide the required background for pursuing research in this area.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*:
	Basics: Introduction. Undirected and Directed Graphical Models. Bayesian Networks. Markov Networks. Exponential Family Models. Factor Graph Representation. Hidden Markov Models. Conditional Random Fields. Triangulation and Chordal Graphs. Other Special Cases: Chains, Trees. Inference: Variable Elimination (Sum Product and Max-Product). Junction Tree Algorithm. Forward Backward Algorithm (for HMMs). Loopy Belief Propagation. Markov Chain Monte Carlo. Metropolis Hastings. Importance Sampling. Gibbs Sampling. Variational Inference. Learning: Discriminative Vs. Generative Learning. Parameter Estimation in Bayesian and Markov Networks. Structure Learning. EM: Handling Missing Data. Applications in Vision, Web/IR, NLP and Biology. Advanced Topics: Statistical Relational Learning, Markov Logic Networks.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Course Outline; Basics	2
2	Directed Graphical Models: Bayesian Networks	4
3	Markov Networks, Factor Graph Representation, Exponential Family Models	4
4	Hidden Markov Models, Conditional Random Fields	3
5	Exact Inference: Variable Elimination	4
6	Exact Inference: Junction Tree Algorithm, Forward Backward Algorithm	4
7	Approximation Inference: MCMC, Metropolis Hastings, Importance Sampling, Gibbs Sampling	6
8	Variational Inference	3
9	Parameter Learning in Bayesian and Markov Networks, Structure Learning	5
10	Handling Missing Data	3
11	Applications and Advanced Topics	4
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities -** NA

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
1	Implementing Bayesian Networks, Markov Networks, Naïve Bayes	6
2	Gibbs Sampling, Loopy Belief Propagation, Junction Tree Algorithm	6
3	Working with Applications (Vision, IR, NLP, Biology)	6
4	Course Project	10
COURSE TOTAL	***(14 times 'P')***	**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Daphne Koller and Nir Friedman. “Probabilistic Graphical Models: Principles and Techniques”. MIT Press, 2009.
2. M. I. Jordan (ed). "Learning in Graphical Models". MIT Press. 1998. Collection of papers.
3. J. Pearl. "Probabilistic Reasoning in Intelligent Systems: Networks of Plausible Inference." Morgan Kaufmann. 1988.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Matlab. Publicly Available Online Datasets (e.g. IR, Vision, Biology etc.).
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Computer Vision
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 780
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	EEL806

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	Prof Subhashis Banerjee
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)***:** This course is meant to be an introduction to the concepts and applications of computer vision. Computer vision, in general deals with methods of extracting and interpreting information from images and videos.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Camera models. Calibration, multi-views projective geometry and invariants. Feature detection, correspondence and tracking. 3D structure/motion estimation. Application of machine learning in object detection and recognition, category discovery, scene and activity interpretation.

15.	**Lecture Outline***(with topics and number of lectures)*	
Module no.	**Topic**	**No. of hours**
1	Camera Models, Calibration and Projective Geometry	6
2	Multi Views Geometry	6
3	Feature Extraction, Correspondence and Tracking	6
4	3D Structure/Motion Estimation	6
5	Applications: Single View Reconstruction, Large Scale Reconstruction	6
6	Object Detection and Recognition	6
7	Scene and Activity Interpretation	6
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities -** NA

17.	**Brief description of laboratory activities** The course structure is 3-0-2, to reinforce some concepts it is desired that students implement and experiment through assignments/projects in the course.

Module no.	**Experiment description**	**No. of hours**
1	Camera Calibration	8
2	Afffine and Metric Rectification	8
3	Project (Open)	12
COURSE TOTAL	***(14 times 'P')***	**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Richard Hatley and Andrew Zisserman, Multiple View Geometry, Cambridge University Press
2. David Forsyth and Jean Ponce, Computer Vision: A Modern Approach
3. Research Papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	Matlab/OpenCV for doing assignments/project.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	Standard
19.4	Laboratory	Standard
19.5	Equipment	Standard
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	30%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: The course is designed to be a mix of theory as well as hands on experience. A background in Linear Algebra and Basic Probability is required.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Computer Graphics
3.	**L-T-P structure**	3-0-3
4.	**Credits**	4.5
5.	**Course number**	COL781
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	COL106

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	MAL754, EEL754 (Old Numbs)

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course**	Subodh Kumar, Prem Kalra, KK Biswas
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To introduce the basic concepts of computer graphics.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Graphics pipeline; Graphics hardware: Display devices, Input devices; Raster Graphics: line and circle drawing algorithms; Windowing and 2D/3D clipping: Cohen and Sutherland line clipping, Cyrus Beck clipping method; 2D and 3D Geometrical Transformations: scaling, translation, rotation, reflection; Viewing Transformations: parallel and perspective projection; Curves and Surfaces: cubic splines, Bezier curves, B-splines, Parametric surfaces, Surface of revolution, Sweep surfaces, Fractal curves and surfaces; Hidden line/surface removal methods; illuminations model; shading: Gouraud, Phong; Introduction to Ray-tracing; Animation; Programming practices with standard graphics libraries like openGL

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Graphics pipeline and hardware	4
2	Raster Graphics	3
3	2D and 3D Clipping	3
4	Transformations	6
5	Curves and Surfaces	9
6	Hidden Surface Elimination	5
7	Illumination Model and Shading	3
8	Ray Tracing	5
9	Animation	4
COURSE TOTAL *(14 times 'L')*		
		42

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	Experiment description	No. of hours
1	OpenGL based application	14
2	Rendering algorithm implementation	14
3	Modeling and animation	14
4		
COURSE TOTAL *(14 times 'P')*		**42**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Foley, J. D., van Dam, A., Feiner, S. K., and Hughes, J. F., Computer Graphics: Principles and Practice, Pearson Addison Wesley, 1990.
2. Hearn, D., and Baker, P. M., Computer Graphics, Pearson Higher Education, 1994.
3. Rogers, F. D., and Adams, J. Alan, Mathematical Elements for Computer Graphics, McGraw Hill, 1999.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	.
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	Projector with A/V
19.4	Laboratory	Yes
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	40
20.4	Open-ended laboratory work	30
20.5	Others (please specify)	30 Implementation

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	DIGITAL IMAGE ANALYSIS
3.	**L-T-P structure**	3-0-3
4.	**Credits**	4.5
5.	**Course number**	COL783
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre Requisites** (course no./title)	EEL205, (Old Numbers) COL106

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	MAL715, EEL715, PHL756 (Old Numbers)

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course**	Prem Kalra, K K Biswas, Subhashis Banerjee
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To introduce basic concepts of image processing.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Digital Image Fundamentals; Image Enhancement in Spatial Domain: Gray Level Transformation, Histogram Processing, Spatial Filters; Image Transforms: Fourier Transform and their properties, Fast Fourier Transform, Other Transforms; Image Enhancement in Frequency Domain; Color Image Processing; Image Warping and Restoration; Image Compression; Image Segmentation: edge detection, Hough transform, region based segmentation; Morphological operators; Representation and Description; Features based matching and Bayes classification; Introduction to some computer vision techniques: Imaging geometry, shape from shading, optical flow; Laboratory exercises will emphasize development and evaluation of image processing methods.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Digital Image Fundamentals	2
2	Image Enhancement in Spatial Domain	3
3	Image Transforms	6
4	Image Enhancement in Frequency Domain	2
5	Color Image Processing	2
6	Image Warping and Restoration	4
7	Image Compression	6
8	Image Segmentation	4
9	Morphological Operators	3
10	Representation and Description	3
11	Image Matching and Classification	4
12	Introduction to some Computer Vision techniques	3
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	Experiment description	No. of hours
1	Image Enhancement in Spatial and Frequency domain	8
2	Hough Transforms	8
3	Image Warping and Restoration	8
4	Image Compression	8
5	Open Assignment	10
COURSE TOTAL ***(14 times 'P')***		**42**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Gonzalez, R. C., and Woods, R. E., Digital Image Processing, Addison-Wesley.
2. Pitas, I., Digital Image Processing - Algorithms and Applications, John Wiley.
3. Sonka, M., Hlavac, V., and Boyale, R., Image Processing, Analysis and Machine Vision,

PWS Publishing (ITP-International Thomson Publishing).
4. Jain, A. K., Fundamentals of Digital Image Processing, Printice Hall of India (PHI).

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	.
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	Projection A/V
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	40
20.4	Open-ended laboratory work	20
20.5	Others (please specify)	40 Implementation

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Functional Brain Imaging
3	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 786
6.	**Status** (*category* for *program*)	PE for MTech (MCS), Dual, CSY/SIY/ANZ/CSZ DE for BTech, Dual (CS); Open Category

7.	**Pre Requisites** (course no./title)	--

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	BML720 -- Medical Imaging (CBME) < 15% overlap.
8.2	Overlap with any UG/PG course of other Dept./Centre	No
8.3	Supersedes any existing course	No

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every year

11.	**Faculty who will teach the course**	Rahul Garg
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: Recent advances in functional Brain and neuroimaging have come from the application of several advanced machine learning techniques. This course aims to cover these. The course will start with an introduction to the human brain anatomy and function. The students will then learn about various imaging technologies for measuring and modeling brain activities. Finally, recent advances, including functional connectivity and applications of machine learning techniques in analyzing the brain data will be covered. The course involve experimentation where students will be able to apply the knowledge gained in the course for the analysis of Neuroimaging data.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Introduction to human Neuroanatomy, Hodgkin Huxley model, overview of brain imaging methods, introduction to magnetic resonance imaging, detailed fMRI, fMRI data analysis methods, general linear model, network analysis, machine learning based methods of analysis.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to Brain and its function	6
2	Basic Human neuroanatomy	6
3	Hodgkin Huxley model of neuronal firing	3
4	Overview of brain imaging methods	3
5	Introduction to MRI	3
6	Introduction to medical image reconstruction	3
7	Functional MRI	3
8	fMRI data analysis – GLM analysis	6
9	fMRI data analysis – network analysis	3
10	Machine learning in brain data analysis	6
COURSE TOTAL	***(14 times 'L')***	**42**

16.	**Brief description of tutorial activities: NA**

17.	**Brief description of laboratory activities:**

Module no.	Experiment description	No. of hours
1	Identify anatomical structures in brain from MRI images	2
2	Analysis of spike train data	2
3	Analysis of EEG/MEG/ECOG data	4
4	MR Image reconstruction using simulator	4
5	Radon transform and solution to inverse problem in a plane	2
6-10	fMRI data analysis	14
COURSE TOTAL	***(14 times 'P')***	28

18.	**Suggested texts and reference materials**

1. Human Brain Function (2004), by Richard S.J. Frackowiak, John T. Ashburner, William D. Penny, Semir Zeki, Karl J. Friston, Christopher D. Frith, Raymond J. Dolan, Cathy J. Price, Academic Press.
2. Principles of Neural Science (2012), by Eric R. Kandel, James H. Schwartz, Thomas M. Jessell, Steven A. Siegelbaum, A. J. Hudspeth, McGraw-Hill Professional

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Neurodebian
19.2	Hardware	Access to computers
19.3	Teaching aides (videos, etc.)	Following brain documentaries will be helpful. 1. Secrets of the Mind by V.S. Ramachandran PBS NOVA Documentary, originally broadcast on

		October 23, 2001 2. The Secret Life of the Brain, PBS Documentary.
19.4	Laboratory	Server with Neurodebian installed
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector, speakers and amplifier for audio
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	10%
20.2	Open-ended problems	15%
20.3	Project-type activity	50%
20.4	Open-ended laboratory work	15%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1	**Department/Centre proposing the course**	Department of Computer Science and Engineering
2	**Course Title** *(< 45 characters)*	Advanced Topics in Embedded Computing
3	**L-T-P structure**	3-0-2
4	**Credits**	4
5	**Course number**	COL 788
3	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ ,VDTT DE for CS1, CS5

7	**Pre-requisites** (course no/title)	COL216,COL331 or equivalent

8	**Status vis-à-vis other courses** *(give course number/title)*	
81	Overlap with any UG/PG course of the Dept/Centre	NIL
82	Overlap with any UG/PG course of another Dept/Centre	EEL705
83	Supersedes any existing course	NA

9	**Not allowed for** (indicate program names)	

10	**Frequency of offering**	Alternative semester

11	**Faculty who will teach the course** 1. Kolin Paul, S R Sarangi, Anshul Kumar	
12	**Will the course require any visiting faculty?**	No

13	**Course objective** *(about 50 words; motivation and aims of this course)*: Modern Embedded Computing systems are used for feature rich highly integrated handheld mobile devices. The embedded platforms use sophisticated processors like Cortex and Atom to deliver performance. Topics ranging from using these embedded platforms to system initialization and debug will be covered. Program optimizations to ensure real time deadlines are met in typical media applications will be covered. The course will be systems oriented encompassing a mix of topics in Processor architecture and Real Time Operating systems.
14	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)*: Embedded Platforms , Embedded processor architectures, System initialization, Embedded operating systems (linux) , DSP and graphics acceleration, Interfaces, Device Drivers, Network, Security, Debug support, Performance tuning. The course would involve substantial programming assignments on embedded platforms.

15	**Lecture Outline** *(with topics and number of lectures)*	
Module no	**Topic**	**No of hours**
1	Embedded Platforms	3
2	Embedded processor architectures	9
3	System initialization	3
4	Embedded operating systems (linux)	9
5	DSP and graphics acceleration	3
3	Network	3
7	Security	3
8	Performance tuning	3
9	Device Drivers	3
10	Advanced Issues	3
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

13	**Brief description of tutorial activities**

17	**Brief description of laboratory activities**	
Module no	**Experiment description**	**No of hours**
1	IDE in the linux environment Hello World	3
2	Boot Process	5
3	DSP	3
4	Multithreaded Programming	3
5	Real Time Performance	3
3	ICE and Debugging	3
7	Performance Tuning	3
8	Code Optimizations	5
	COURSE TOTAL *(14 times 'P')*	28

18	**Suggested texts and reference materials**
	1. Modern Embedded Computing: Designing Connected, Pervasive, Media-Rich Systems ISBN-10: 0123914903

19	**Resources required for the course** *(itemized & student access requirements, if any)*	
191	Software	Open Source
192	Hardware	
193	Teaching aides (videos, etc)	
194	Laboratory	
195	Equipment	Hardware kits
193	Classroom infrastructure	
197	Site visits	
20	**Design content of the course** *(Percent of student time with examples, if possible)*	
201	Design-type problems	25
202	Open-ended problems	05
203	Project-type activity	45
204	Open-ended laboratory work	15
205	Others (Crtique Research Papers)	10

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	System Level Design and Modelling
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 812
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL719

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐Every sem ☐1st sem ☒2nd sem ☐Alternate year
11.	Faculty who will teach the course	1. M Balakrishnan, P R Panda
12.	**Will the course require any visiting faculty?**	No
13.	**Course objective** *(about 50 words)*: To understand the principles of modelling and design of complex embedded systems with both hardware and software components.	
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Embedded systems and system-level design, models of computation, specification languages, hardware/software co-design, system partitioning, application specific processors and memory, low power design.	

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to embedded systems	4
2	Models of computation	4
3	Specification languages	3
4	Behavioural synthesis review	4
5	Hardware/software codesign	5
6	Application specific processors	4
7	Application specific memory	4
8	Low power design	4
9	Special topics and recent developments	10
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Modelling and simulation of a reasonably complex embedded system application.	
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Michel, P., Lauther, U., and Duzy, P. (eds.), The synthesis approach to digital systems, Kluwer Academic Publishers, Norwell, 1992.
2. Gajski, D. D., Vahid, F., Narayan, S., and Gong, J., Specification and Design of Embedded Systems, Prentice Hall, New Jersey, 1994
3. Lee, E. A., and Parks, T. M., Dataflow Process Networks, Proceedings of the IEEE, May 1995, pg 773-801
4. SystemC Version 2.0 User's Guide
5. Smith, M. J. S., Application -Specific Integrated Circuits, Pearson Education,1997 (Chapter 15, pg 824-838)
6. Jain, M. K., Balakrishnan, M., and Kumar, A., ASIP Design Methodologies: Survey and Issues, International Conference on VLSI Design, 2001

7. Panda, P. R., Dutt, N. D., and Nicolau, A., Memory issues in embedded systems-on-chip: optimizations and exploration, Kluwer Academic Press, 1999.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Department of Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Principles of Multiprocessor Systems
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 818
6.	**Status** (*category* for *program*)	PE for MTech (MCS), CS5 (Dual), CSY/SIY/ANZ/CSZ DE for CS1, CS5

7.	**Pre-requisites** (course no./title)	COL 216: Computer Architecture COL 351: Analysis and Design of Algorithms COL 331: Operating Systems

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	10% overlap with CSL 718 (advanced architectures) and CSL 730 (parallel programming)
8.2	Overlap with any UG/PG course of another Dept./Centre	No
8.3	Supersedes any existing course	NA

9.	**Not allowed for** (indicate program names)	NA

10.	**Frequency of offering**	Alternate semester

11.	**Faculty who will teach the course** 1. Dr. Smruti Ranjan Sarangi 2. Prof. S. Arun Kumar 3. Prof. Sanjiva Prasad	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words; motivation and aims of this course)*: This course will give graduate students a thorough understanding of the major concepts underlying the software and hardware aspects of shared memory multiprocessor systems. It will also expose them to the latest technologies in this rapidly emerging area.
14.	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)*: Mutual Exclusion, Coherence and Consistency, Register Constructions , Power of Synchronization Operations , Locks and Monitors, Concurrent queues, Futures and Work-Stealing, Barriers, Basics of Transactional Memory (TM), Regular Hardware TMs, Unbounded HadwareTMs, Software TMs

15.	**Lecture Outline** *(with topics and number of lectures)*	
Module no.	**Topic**	**No. of hours**
1	Introduction	1
2	Mutual Exclusion	3
3	Coherence and Consistency	4
4	Register Constructions	1.5
5	Power of Synchronization Operations	3
6	Universal Construction	2
7	Locks	3
8	Monitors	1.5
9	Concurrent Queues	3
10	Futures and Work Stealing	4
11	Barriers	3
12	Basics of Transactional Memory	3
13	Regular Hardware Transactional Memory	3
14	Unbounded Hardware Transactional Memory	1.5
15	Opacity and Fo-consensus	2.5
16	Software Transactional Memory	3
	COURSE TOTAL *(14 times 'L')*	**42**

16.	**Brief description of tutorial activities - NA**	
17.	**Brief description of laboratory activities**	
Module no.	**Experiment description**	**No. of hours**
1	Design of mutual exclusion primitives	5
2	Implementation of non-blocking data structures	5
3	A parallel future based work scheduler	6
4	Writing programs using software transactional memory frameworks	6
5	Implementation of a minimal STM	6
	COURSE TOTAL *(14 times 'P')*	**28**

18.	**Suggested texts and reference materials** 1. Art of Multiprocessor Programming, Herlihy and Shavit, 1st edition, 2009. Morgan Kaufmann Publication. 2. Principles of Transactional Memory, Guearroui and Kapalka, 1st edition, 2011. Morgan and Claypool Publications	
19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	C++ and Java compilers
19.2	Hardware	Parallel computer with at least 16 cores. They are already available in the CSE and SIT deptts.
19.3	Teaching aides (videos, etc.)	No
19.4	Laboratory	No
19.5	Equipment	No
19.6	Classroom infrastructure	None
19.7	Site visits	None
20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	20.00%
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Department of Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Distributed Systems
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 819
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL 331 (Operating systems), COL 334 (Computer networks), COL 380 (Introduction to parallel and distributed computing)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	CSL 830 (Distributed Computing) – 10% overlap, CSL 847(Distributed Algorithms) – 15% overlap
8.2	Overlap with any UG/PG course of another Dept./Centre	NA
8.3	Supersedes any existing course	NA

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every Alternate Semester

11.	**Faculty who will teach the course** 1. Smruti Ranjan Sarangi 2. Sanjiva Prasad 3. Sorav Bansal

12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words; motivation and aims of this course)***:** The course has broadly three objectives – undestanding of the basic paradigms in large scale distributed systems, basic distributed algorithms, and a description of real life case studies.
14.	**Course contents** *(about 100 words; this is a topics wise listing that will appear in the Courses of Study) (Include laboratory/design activities)***:** Epidemic/Gossip based algorithms, Peer to peer networks, Distributed hash tables, Synchronization, Mutual exclusion, Leader election, Distributed fault tolerance, Large scale storage systems, Distributed file systems, Design of social networking systems

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction	3
2	Epidemic based algorithms	3
3	Gossip based algorithms	2
4	Peer to peer systems	7
5	Synchronization and logical clocks	3
6	Consistency and replication	3
7	Mutual exclusion and leader election	2
8	Fault Tolerance	5
9	Distributed storage systems	4
10	Distributed file systems	3
11	Publish subscribe systems	2
12	Design of social networking platforms	5
Total		42

16. Brief description of tutorial activities - NA

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Implemention of a distributed hash table	8
2	Implementation of a distributed algoritm. Examples: Finding the minimum spanning tree, leader election, mutual exclusion	8
3	Implement a distributed file system	6
4	Implementation of a publish subscribe system	6
	COURSE TOTAL ***(14 times 'P')***	28

18.	**Suggested texts and reference materials** 1. Distributed Systems (Tanenbaum and Marten V. Steen) 2. Distributed Algorithms (Gerard Tel) 3. Distributed Algorithms (Nancy Lynch) 4. Introduction to Reliable and Secure Distributed Programming (Rachid Guerraoui)	
19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	Open source software
19.2	Hardware	24 core servers, present in the CSE department
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	
20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	30
20.2	Open-ended problems	30
20.3	Project-type activity	20
20.4	Open-ended laboratory work	20
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Reconfigurable Computing
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 821
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL719

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐Every sem ☒1st sem ☐2nd sem ☒Alternate year
11.	**Faculty who will teach the course**	Kolin Paul, Anshul Kumar
12.	**Will the course require any visiting faculty?**	No

13. Course objective *(about 50 words)*:

The objective of this course is to familiarize the students with the current research on various aspects of reconfigurable computing, including reconfigurable devices, architecture and compilers. The course will involve a fair amount of critically analyzing a digest of papers. There will also be a semester long implementation project.

14. Course contents *(about 100 words) (Include laboratory/design activities)*:

	FPGA architectures, CAD for FPGAs: overview, LUT mapping, timing analysis, placement and routing, Reconfigurable devices - from fine-grained to coarse-grained devices, Reconfiguration modes and multi-context devices, Dynamic reconfiguration, Compilation from high level languages, System level design for reconfigurable systems: heuristic temporal partitioning and ILP-based temporal partitioning, Behavioral synthesis, Reconfigurable example systems' tool chains

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction to Reconfigurable Computing	1
2	Designing for FPGA	2
3	Logic Cells, Tech Mapping and Placement	3
4	CAD of FPGAs - a) the Xilinx tool chain, b) building systems with HandelC	3
5	Systolic architectures and applications	6
6	Systolic loop transformations - a) dependence analysis, b) scheduling, c) localization	6
7	Software pipelining	3
8	Partial reconfiguration, multi-context architectures	3
9	Runtime reconfigurable architectures, Piperench	3
10	Behavioral synthesis	6
11	Example reconfigurable systems' tool chains - COMPAAN (with KPNs), StreamsC (with CSP), MMAlpha	6
12		
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	There will be a semester long project as a part of the course	
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

Scott Hauck and Andre DeHon, *Reconfigurable Computing: The Theory and Practice of FPGA-Based Computation (Systems on Silicon)* Morgan Kaufmann 2007

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	SPECIAL TOPICS IN HARDWARE SYSTEMS
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL861
6.	**Status** (*category* for *program*)	PE for **CO**

7.	**Pre-requisites** (course no./title)	E.C. 120

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year

11.	**Faculty who will teach the course** Anshul Kumar, M. Balakrishnan, Preeti R Panda	
12.	**Will the course require any visiting faculty?**	No
13.	**Course objective** *(about 50 words)*:	
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Under this topic one of the following areas will be covered: Fault Detection and Diagnosability. Special Architectures. Design Automation Issues. Computer Arithmetic, VLSI	

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
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6		
7		
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9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	INDEPENDENT STUDY
3.	**L-T-P structure**	0-3-0
4.	**Credits**	3
5.	**Course number**	COS310
6.	**Status** (*category* for *program*)	DE for **CS,** DE for **CO**

7.	**Pre-requisites** (course no./title)	EC 60

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	non-CS students
10.	**Frequency of offering**	☒Every sem ☐1st sem ☐2nd sem ☐Alternate year
11.	Faculty who will teach the course	All
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Research oriented activities or study of subjects outside regular course offerings under the guidance of a faculty member. Prior to registration, a detailed plan of work should be submitted by the student to the Head of the Department for approval
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Research oriented activities or study of subjects outside regular course offerings under the guidance of a faculty member. Prior to registration, a detailed plan of work should be submitted by the student to the Head of the Department for approval

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
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11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

Practical Training (v1)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	SPECIAL MODULE IN HARDWARE SYSTEMS
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV881
6.	**Status** (*category* for *program*)	PE for **CO**

7.	**Pre-requisites** (course no./title)	Permission of Instructor

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year
11.	Faculty who will teach the course	
12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special module that focuses on special topics and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
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6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times ‘L’)*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times ‘P’)*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Advanced Computer Graphics
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL829
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL781 or Equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course**	Prem Kalra, Subodh Kumar
12.	**Will the course require any visiting faculty?**	NO

13.	**Course objective** *(about 50 words)*: To provide some advance concepts in modeling, rendering and animation.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Rendering: Ray tracing, Radiosity methods, Global illumination models, Shadow generation, Mapping, Anti-aliasing, Volume rendering, Geometrical Modeling: Parametric surfaces, Implicit surfaces, Meshes, Animation: spline driven, quarternions, articulated structures (forward and inverse kinematics), deformation -- purely geometric, physically-based, Other advanced topics selected from research papers.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Ray Tracing	6
2	Radiosity	4
3	Global Illumination	3
4	Shadows and Mapping	4
5	Volume Rendering	3
6	Geometrical Modeling	6
7	Animation	8
8	Other topics from papers	8
9		
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	Experiment description	No. of hours
1	Project involving advanced concepts of modeling, rendering and animation	
2		
3		
4		
COURSE TOTAL ***(14 times 'P')***		**28**

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Watt, A., and Watt, M., Advanced Animation and Rendering Techniques, Theory and Practice, Addison Wesley, 1992.
2. Research Papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	.
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	Projection with A/V
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	50
20.4	Open-ended laboratory work	30
20.5	Others (please specify)	20 Implementation

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Distributed Computing
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 830
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL226

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Alternate year

11.	**Faculty who will teach the course** Sanjiva Prasad, S. Arun Kumar, S.N. Maheshwari

12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To introduce the practical distributed computing problems within the framework of abstract mathematical models, and develop the tools and techniques using which non-trivial properties can be identified, their complexity assessed and distributed systems specified.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Models of Distributed Computing; Basic Issues: Causality, Exclusion, Fairness, Independence, Consistency; Specification of Distributed Systems: Transition systems, petri nets, process algebra properties: Safety, Liveness, stability.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Models of Distributed Computing	5
2	Basic issues	7
3	Synchrony	8
4	Fault-Tolerance	8
5	Specification and Properties	10
6	Applications and Practical Exercises	4
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
	COURSE TOTAL *(14 times 'P')*	

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Lynch, N. A., Distributed Algorithms, Morgan Kaufmann, 1996.
2. Gerald T., Topics in Distributed Algorithms, Cambridge University Press, 1991
3. Papers published in current journals.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Semantics of Programming Languages
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 831
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL226, COL352

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Alternate year
11.	**Faculty who will teach the course** Sanjiva Prasad, S. Arun Kumar	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To introduce the important results in the area of programming language semantics, and the necessary mathematical frameworks for reasoning about programs.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Study of operational, axiomatic and denotational semantics of procedural languages; semantics issues in the design of functional and logic programming languages, study of abstract data types.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours

1	Syntax and Semantics Foundations	3
2	Operational Semantics	5
3	Denotational Semantics	5
4	Algebra of Data Types and Category Theory	6
5	Program Logics, semantics, correctness	6
6	Recursion in programs (fixpoints)	5
7	Solving recursive domain equations	8
8	Lambda Calculus and Domains	4
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities - NA

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Using theorem proving techniques to implement semantics definitions and prove correctness of meta theorems.	
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Winskel, G. The Formal Semantics of Programming Languages, MIT Press, 1993.
2. Gunter, C. A., Semantics of Programming Languages: Structures and Techniques, MIT Press, 1992.
3. Reynolds, J., Theories of Programming Languages, Cambridge Univ Press, 1998.
4. Watt, D. A., Programming Concepts and Paradigms, Prentice-Hall, 1990.
5. Stoy, J., Denotational Semantics, MIT Press 1977.
6. Schmidt, D. A., Denotational Semantics, Allyn and Bacon, 1986.
7. Various notes and journal material (Plotkin's notes on Operational Semantics and Denotational Semantics, Domain Theory,...)

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	

20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Proofs and Types
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 832
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL226, COL352

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Alternate year

11.	**Faculty who will teach the course** Sanjiva Prasad, S. Arun KUmar	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course aims to introduce the deep connection between logic, programming languages, computation theory and the foundations of theorem proving.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Syntax and semantic foundations: Ranked algebras, homomorphisms, initial algebras, congruences. First-order logic review: Soundness, completeness, compactness. Herbrand models and Herbrand's theorem, Horn-clauses and resolution. Natural deduction and the Sequent calculus. Normalization and cut elimination. Lambda-calculus and Combinatory Logic: syntax and operational semantics (beta-eta equivalence), confluence and Church-Rosser property. Introduction to Type theory: The simply-typed lambda-calculus, Intuitionistic type theory. Curry-Howard correspondence. Polymorphism, algorithms for polymorphic type inference, Girard and Reynolds' System F. Applications: type-systems for programming languages; modules and functors; theorem proving, executable specifications.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Syntax and Semantics: Algebras, terms, homomorphisms	3
2	Review of First order logic, Set Theory	4
3	Natural Deduction and sequent calculus	4
4	Lambda calculus and simply typed Lambda calculus	5
5	Normalization, Reducibility. Curry-Howard isomorphism	6
6	Church's Theory of Types and Inductive types	7
7	Dependent Types and applications	6
8	Polymorphism, System F and Calculus of Constructions	7
9		
10		
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1	Unification, Backtracking, BFS, DFS Environment machines for Lambda calculus(SECD,Krivine)	
2	Graph reduction/combinatory reduction.	
3	Polymorphic type-checking.	
4	Theorem Proving using Coq or Isabelle/HOL.	
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	COURSE TOTAL *(14 times 'P')*	

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Girard, J., Lafont Y., and Taylor, P., Proofs and Types, Cambridge University Press, 1989
2. Hindely, J. R., and Seldin, J. P., Introduction to Combinators and Lambda Calculus, Cambridge University Press, 1989
3. Dowek, Huet, Paulin, Types and Proofs, 1995, downloadable.
4. The Isabelle Proof Assistant, Paulson and Nipkow, Springer, 2001, downloadable.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	Designing type systems for programming languages
20.2	Open-ended problems	applications of linear type systems
20.3	Project-type activity	Verification of theorems
20.4	Open-ended laboratory work	implementation of type systems
20.5	Others (please specify)	Paper presentations, generic programming techniques

Date: (Signature of the Head of the Department)

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COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Operating Systems
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL851
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	OPERATING SYSTEMS

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	Frequency of offering	Every other sem

11.	**Faculty who will teach the course** Huzur Saran, Sanjiva Prasad, S. Arun Kumar, Sorav Bansal, S.C. Gupta, Subodh Kumar, Kolin Paul, Smruti Ranjan Sarangi, Subhashis Banerjee	
12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in the area of operating systems. Topics may include, but are not limited to, OS design, web servers, Networking stack, Virtualization, Cloud Computing, Distributed Computing, Parallel Computing, Heterogeneous Computing, etc.

-

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special topic that focuses on state of the art and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Lectures contents will depend on the exact topics being covered. They will typically involve discussing recent and classic research papers in the area	
2		
3		
4		
5		
6		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Moduleno.	Experiment description	No. of hours
1		
2		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

Research papers, online material. Perhaps, some advanced books, but that will depend on the topics being covered.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	Usually Open-source Software
19.2	Hardware	One shared lab computer per student
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

-

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	15%
20.2	Open-ended problems	15%
20.3	Project-type activity	15%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

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COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in COMPILER DESIGN
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL852
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	Frequency of offering	Every other sem

11.	**Faculty who will teach the course**	Sanjiva Prasad, S. Arun Kumar, Sorav Bansal
12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in the area of compiler design. Topics may include, but are not limited to, type systems, language design, compiler design, compiler optimization, code generation, binary translation, etc.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special topic that focuses on state of the art and research problems of importance in this area

-

15. **Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Lectures contents will depend on the exact topics being covered. They will typically involve discussing recent and classic research papers in the area	
2		
3		
4		
5		
6		
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8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

Research papers, online material. Perhaps, some advanced books, but that will depend on the topics being covered.

-

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	Usually Open-source Software
19.2	Hardware	One shared lab computer per student
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	15%
20.2	Open-ended problems	15%
20.3	Project-type activity	15%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Parallel Computation
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL860
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course**	Subodh Kumar, Rahul Garg, Sandeep Sen
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is aimed at providing students with a deep knowledge of the techniques and tools needed to understand today's and tomorrow's high performance computers, and to efficiently program them.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The course will focus on research issues in areas like parallel computation models, parallel algorithms, Parallel Computer architectures and interconnection networks, Shared memory parallel architectures and programming with OpenMP and Ptheards, Distributed memory message-passing parallel architectures and programming, portable parallel message-passing programming using MPI. This will also include design and implementation of parallel numerical and non-numerical algorithms for scientific and engineering, and commercial applications. Performance evaluation and benchmarking high-performance computers.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Parallel Computer architectures and interconnection networks	4
2	Shared memory parallel architectures and programming with OpenMP and Ptheards	4
3	Distributed memory message-passing parallel architectures and programming	6
4	portable parallel message-passing programming using MPI	6
5	Performance evaluation and benchmarking high-performance computers	6
6	parallel numerical and non-numerical algorithms	6
7	This will also include design and implementation seminars	10
8		
9		
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	**Experiment description**	**No. of hours**
1		
2		
3		
4		
COURSE TOTAL ***(14 times 'P')***		

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Research Papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	.
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Harware Systems
3.	**L-T-P structure**	3-0-2
4.	**Credits**	4
5.	**Course number**	COL 861
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Alternate year

11.	**Faculty who will teach the course** Anshul Kumar, M. Balakrishnan, Preeti R Panda	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*:

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Under this topic one of the following areas will be covered: Fault Detection and Diagnosability. Special Architectures. Design Automation Issues. Computer Arithmetic, VLSI

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2	Lectures and student seminars	42
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Moduleno.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	

19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Software Systems
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 862
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Alternate year
11.	**Faculty who will teach the course** Rahul Garg, Sorav	
12.	**Will the course require any visiting faculty?**	May be

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special topic that focuses on state of the art and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5^{th} Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Theoretical Computer Science
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 863
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL 351 (Analysis and Algorithm Design) or equivalent

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	--

10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year

11.	**Faculty who will teach the course** Sandeep Sen, Amit Kumar, Naveen Garg, SN Maheshwari, Ragesh Jaiswal	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To explore new and recent trends in theoretical computer science.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Under this topic one of the following areas will be covered: Design and Analysis of Sequential and Parallel Algorithms. Complexity issues, Trends in Computer Science Logic, Quantum Computing and Bioinformatics, Theory of computability. Formal Languages. Semantics and Verification issues.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2	Lectures and student seminars	42 42
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
	COURSE TOTAL *(14 times 'L')*	42

16. Brief description of tutorial activities

NA

17. Brief description of laboratory activities

Modul eno.	Experiment description	No. of hours
1		
2		
3		
4		
5		
	COURSE TOTAL *(14 times 'P')*	

18.	**Suggested texts and reference materials** 1. Research papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Artificial Intelligence
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 864
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL333 or COL671

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No

9.	**Not allowed for**(indicate program names)	---

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	Mausam, Saroj Kaushik, K K Biswas, Parag Singla
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The students will learn about topics, which are currently at the forefront of Artificial Intelligence research. Each semester, the theme of the course may change depending on the instructor.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Potential topics or themes which may be covered (one topic per offering) include: information extraction, industrial applications of AI, advanced logic-based AI, Markov Decision Processes, statistical relational learning, etc.

15.	**Lecture Outline***(with topics and number of lectures)*: No specific list of lectures can be specified since the topic of the course changes from offering to offering.

16.	**Brief description of tutorial activities**: NA

17.	**Brief description of laboratory activities:** NA

18.	**Suggested texts and reference materials**

1. Papers published in conferences and journals will be suggested based on the topic.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	NA
19.2	Hardware	NA
19.3	Teaching aides (videos, etc.)	NA
19.4	Laboratory	NA
19.5	Equipment	NA
19.6	Classroom infrastructure	Projector
19.7	Site visits	N.A.

20.	**Design content of the course***(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	Writing reports, presenting papers, potentially around 50%

20.4	Open-ended laboratory work	
20.5	Others (please specify)	Reading papers, potentially around 50%

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Computer Applications
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL865
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	Permission of Instructor

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Alternate year

11.	**Faculty who will teach the course** Rahul Garg	
12.	**Will the course require any visiting faculty?**	May be

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area of Applications of Computer Science. The exact area of computer applications may differ every year. Topics may include applications in Medical & Brain Imaging, Education and Massive Online Learning etc.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special topic that focuses on special topics and research problems of importance in this area. These may include applications in Neuroimaging, MOOC education etc.

15. Lecture Outline *(with topics and number of lectures)*

Modul e no.	Topic	No. of hours
1		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Modul eno.	Experiment description	No. of hours
1		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Research papers

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Algorithm
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 866
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL 351 (Analysis and Algorithm Design) or equivalent

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	No
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year
11.	**Faculty who will teach the course** Sandeep Sen, Amit Kumar, Naveen Garg, SN Maheshwari, Ragesh Jaiswal	
12.	**Will the course require any visiting faculty?**	No

13. Course objective *(about 50 words)*:

Second level advanced course in algorithms for students especially at the pre Ph.D. level. The faculty can also cover recent developments in the area of algorithms that have not yet found their way into text books.

14. Course contents *(about 100 words) (Include laboratory/design activities)*:

The course will focus on specialized topics in areas like Computational Topology, Manufacturing processes, Quantum Computing, Computational Biology, Randomized algorithms and other research intensive topics.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
10		
COURSE TOTAL *(14 times 'P')*		

18.	**Suggested texts and reference materials-** Research Papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*	
20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in High Speed Networks
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 867
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL334, COL672

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	N/A
8.2	Overlap with any UG/PG course of other Dept./Centre	N/A

9.	**Not allowed for** (indicate program names)	N/A

10.	**Frequency of offering**	Alternate semester

11.	**Faculty who will teach the course**	1. V. Ribeiro, 2. Huzur Saran, 3. A. Seth
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Computer Networks is a rapidly evolving field. The course will introduce students to advanced topics in Networks and open up current research problems in these areas.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The course will be delivered through a mix of lectures and paper reading seminars on advanced topics in Computer Networks. Hands-on projects will be conceptualized to challenge students to take up current research problems in areas such as software defined networking, content distribution, advanced TCP methodologies, delay tolerant networking, data center networking, home networking, green networking, clean state architecture for the Internet, Internet of things, etc.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Crash course on computer networks	4
2	Relevant basics such as scheduling, flow control, congestion control, error detection, reliability, switching, routing, addressing	6
3	Relevant background landscape such as design principles of the Internet, network stack	5
4	Modules for the chosen advanced area	22
5	Periodic review of project activities	5
COURSE TOTAL ***(14 times 'L')***		**42**

16.	**Brief description of tutorial activities:** NA

17.	**Brief description of laboratory activities:** NA

18.	**Suggested texts and reference materials**

Papers from conferences such as SIGCOMM, INFOCOM, HOTNETS, IMC, PAM, MOBICOM, MOBISYS, NSDI, CCR, CONEXT.

Books: 1. Top-down approach to computer networks, by Kurose and Ross 2. Computer networks: A systems approach, by Peterson and Davie 3. Engineering approach to computer networks, by S. Keshav 4. Patterns in network architecture, by John Day

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Paper reading, review, and presentation	40%
20.2	Open-ended project	40%
20.3	Assignments	20%

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Database Systems
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 868
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL362

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No

9.	**Not allowed for**(indicate program names)	---
10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	S.K. Gupta, Maya Ramanath
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)***:** The students will learn about topics, which are currently at the forefront of Database and Data Management research. Each semester, the theme of the course may change depending on the instructor.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)***:** Potential topics or themes which may be covered (one topic per offering) include: data mining, big data management, information retrieval and database systems, semantic web data management, etc.

15.	**Lecture Outline** *(with topics and number of lectures)*: No specific list of lectures can be specified since the topic of the course changes from offering to offering.

16.	**Brief description of tutorial activities**: NA

17.	**Brief description of laboratory activities**: NA

18.	**Suggested texts and reference materials**

1. Papers published in conferences and journals will be suggested based on the topic.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*	
19.1	Software	NA
19.2	Hardware	NA
19.3	Teaching aides (videos, etc.)	NA
19.4	Laboratory	NA
19.5	Equipment	NA
19.6	Classroom infrastructure	Projector
19.7	Site visits	N.A.
20.	**Design content of the course** *(Percent of student time with examples, if possible)*	

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	Writing reports, presenting papers, potentially around 50%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	Reading papers, potentially around 50%

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Concurrency
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 869
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course** Sanjiva Prasad, S. Arun Kumar	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Make students familiar with the key notions of concurrency theory, models of concurrent systems, Specificational methods, Tools and techniques.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The course will focus on research issues in concurrent, distributed and mobile computations. Some of the following topics will be covered: Models of Concurrent, Distributed and Mobile computation. Process calculi, Event Structures, Petri Nets an labeled transition systems. Implementations of concurrent and mobile, distributed programming languages. Logics and specification models for concurrent and mobile systems. Verification techniques and algorithms for model checking. Type systems for concurrent/mobile programming languages. Applications of the above models and techniques.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction	1
2	Transition systems, Petri Nets	6
3	Notions of observation Behavioural relations	6
4	Liveness and safety: Deadlocks, Divergence, fairness	3
5	Automata-theoretic techniques	9
6	Process algebras	9
7	Logics for concurrency	8
COURSE TOTAL *(14 times 'L')*		42

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5^{th} Ed., John Wiley, 2000.

Various recent books on concurrent systems, papers and materials available on the web.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	15%
20.2	Open-ended problems	15%
20.3	Project-type activity	15%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Machine Learning
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 870
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites**(course no./title)	--

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	

9.	**Not allowed for** (Indicate program names)	

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	K K Biswas, Subhashis Banerjee, Parag Singla, Mausam, Manik Varma
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be a Special Topics course in Machine Learning. The objective of this course is to develop an in depth knowledge/understanding of a particular sub-area of Machine Learning. A set of advanced topics in Machine Learning would be covered depending on instructor's expertise and interests. The course is also aimed to impart skills for carrying out research in the specific sub-area of Machine Learning.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Contents may vary based on the instructor's expertise and interests within the broader area of Machine Learning. Example topics include (but not limiting to) Statistical Relational Learning, Markov Logic, Multiple Kernel Learning, Multi-agent Systems, Multi-Class Multi-label Learning, Deep Learning, Sum-Product Networks, Active and Semi-supervised Learning, Reinforcement Learning, Dealing with Very High-Dimensional Data, Learning with Streaming Data, Learning under Distributed Architecture

15.	**Lecture Outline***(with topics and number of lectures)* Exact Outline will depend on the instructor and the specific content of the course being offered.

16.	**Brief description of tutorial activities -** NA

17.	**Brief description of laboratory activities -** NA

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Kevin P. Murphy. "Machine Learning: A Probabilistic Perspective". MIT Press, 2012.
2. Christopher Bishop. "Pattern Recognition and Machine Learning", First Edition,Springer, 2006.
3. Daphne Koller and Nir Friedman. "Probabilistic Graphical Models: Principles and Techniques". MIT Press, 2009.
4. R. Duda, P. Hart and D. Stork. "Pattern Classification" Second Edition, John Wiley & Sons INC., 2001.
5. Tom Mitchell. "Machine Learning" First Edition, McGraw-Hill, 1997.
6. Relevant Research Papers
7. Online Reading Material.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download (e.g. Weka machine learning repository, UCI datasets). Matlab.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the Course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	40%
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Topics in Programming Languages and Compilers
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 871
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL728, COL729 or equivalent Compilers Course

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No

9.	**Not allowed for** (Indicate program names)	N.A

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	Sanjiva Prasad, P R Panda, SoravBansal
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be a Special Topics course in Programming Languages and Compilers. It will typically cover research-level topics in this area.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Contents may vary based on the instructor's interests within the broader area of Programming Languages and Compilers

15.	**Lecture Outline***(with topics and number of lectures)* Exact Outline will depend on the instructor and the specific content of the course being offered.

16.	**Brief description of tutorial activities** - NA

17.	**Brief description of laboratory activities** N.A.

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Research papers from POPL, PLDI, TOPLAS, etc.
2. Online Reading Material.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	N.A
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the Course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	50%
20.2	Open-ended problems	50%
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** ***(< 45 characters)***	Special Topics in Cryptography
3.	**L-T-P structure**	3-0-0
4.	**Credits**	3
5.	**Course number**	COL 872
6.	**Status (*category* for *program*)**	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites (course no./title)**	COL 759 - Cryptography & Computer Security or Equivalent

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No
8.3	Supersedes any existing course	No

9.	**Not allowed for**(Indicate program names)	NA

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	Shweta Agrawal, Ragesh Jaiswal
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be a Special Topics course in Cryptography. The objective of this course is to develop an in depth knowledge and understanding of new results and current trends in the field. A set of advanced topics would be covered. The course is also aimed to impart skills for carrying out research in the area of Cryptography.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Contents may vary based on the instructor's interests within the broader area of Cryptography. Examples include CCA secure encryption, multiparty computation, leakage resilient cryptography, broadcast encryption, fully homomorphic encryption, obfuscation, functional encryption, zero knowledge, private information retrieval, byzantine agreement, cryptography against extreme attacks etc.

15.	**Lecture Outline***(with topics and number of lectures)* Exact Outline will depend on the instructor and the specific content of the course being offered.

16.	**Brief description of tutorial activities** - NA
17.	**Brief description of laboratory activities -** NA

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.
1. Foundations of Cryptography I and II by OdedGoldreich. Cambridge University Press, 2001. ISBN 0-521-79172-3. 2. Online Reading Material.	

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	N.A
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the Course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	50%
20.2	Open-ended problems	50%
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Software Systems Laboratory
3.	**L-T-P structure**	0-0-6
4.	**Credits**	3
5.	**Course number**	COP701
6.	**Status** (*category* for *program*)	PC for MTech (MCS),

7.	**Pre Requisites** (course no./title)	None

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	Nil
8.2	Overlap with any UG/PG course of other Dept./Centre	Nil

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	1rst sem

11.	**Faculty who will teach the course**	Huzur Saran, Subodh Kumar, SmrutiSarangi, SoravBansal, Kolin Paul, PreetiRanjan Panda, PremKalra, Anshul Kumar, M. Balakrishnan, ParagSingla, almost all CSE faculty members
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To introduce the students to programming abstractions, and make them build fairly large software projects involving for example, GUI creation, distributed state maintenance over a network, programming in different environments like desktop and handhelds, program parsing and compilation, etc.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The contents may differ each year depending on the instructor. The course should involve 2-3 large programming projects done in groups of 2-4.A set of three project oriented assignments which will be announced at the start of each semester with definite submission deadlines. The set of assignments will be designed to develop skills and familarity with a majority of the following: make, configuration management tools, installation of software, archiving and creation of libraries, version control systems, documentation and literate programming systems, GUI creation, distributed state maintenance over a network, programming in different environments like desktop and handhelds, program parsing and compilation including usage of standard libraries like pthreads, numerical packages, XML and semi-structured data, simulation environments, testing and validation tools.

15.	**Lecture Outline***(with topics and number of lectures)* - NA

16.	**Brief description of tutorial activities** - NA

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
1	Project on building GUI, performing distributed synchronization over a network (e.g., game)	40
2	E.g., Project on parsing a programming language and writing a simulator to interpret it. Or building an architectural simulator, etc.	30
3	Any other project (or sub-part of any of the previous projects)	14
COURSE TOTAL *(14 times 'P')*		84

18.	**Suggested texts and reference materials** Programming books, as may be prescribed by the instructor. E.g. C / Java tutorials

1. Project specific

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	GNU/Linux :primarily free and open source software
19.2	Hardware	Commodity desktop, laptop and server machines
19.3	Teaching aides (videos, etc.)	Course webpage
19.4	Laboratory	Enough desktop terminals with all required software installed on them.
19.5	Equipment	None
19.6	Classroom infrastructure	Data projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	30%
20.2	Open-ended problems	
20.3	Project-type activity	70%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Minor Project
3.	**L-T-P structure**	0-0-6
4.	**Credits**	3
5.	**Course number**	COP 891
6.	**Status** (*category* for *program*)	PC for MTech (MCS), Dual (CS)

7.	**Pre Requisites** (course no./title)	--

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	Nil
8.2	Overlap with any UG/PG course of other Dept./Centre	Nil

9.	**Not allowed for** (indicate program names)	--

10.	**Frequency of offering**	Every Sem

11.	**Faculty who will teach the course**	All CSE faculty members
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is a Minor Project for MTech and Dual (CS) Degree Students to work towards their thesis project in the semester before start of the Major Project.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*:

15.	**Lecture Outline** *(with topics and number of lectures)* - NA

16.	**Brief description of tutorial activities** - NA

17.	**Brief description of laboratory activities**

Module no.	Experiment description	No. of hours
COURSE TOTAL ***(14 times 'P')***		84

18.	**Suggested texts and reference materials** Material as may be prescribed by the Supervisor.

1. Project specific

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	30%
20.2	Open-ended problems	
20.3	Project-type activity	70%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Major Project Part 1
3.	**L-T-P structure**	0-0-14
4.	**Credits**	7
5.	**Course number**	COP 892
6.	**Status** (*category* for *program*)	PC for MTech (MCS) and Dual (CS)

7.	**Pre-requisites** (course no./title)	--

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Every sem
11.	Faculty who will teach the course - All	
12.	**Will the course require any visiting faculty?**	May be

13.	**Course objective** *(about 50 words)*: Each student takes up a challenging problem (preferably research-level or a real-life application) and obtains a solution for the same. Each student works on the project under the supervision of any faculty member(s) the Dept. approves.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: It is expected that the problem specification and milestones to be achieved in solving the problem are clearly specified. Survey of the related area should be completed. This project spans also the course CSD852. Hence it is expected that the problem specification and the milestones to be achieved in solving the problem are clearly specified.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5^{th} Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Major Project Part 2
3.	**L-T-P structure**	0-0-28
4.	**Credits**	14
5.	**Course number**	COP 893
6.	**Status** (*category* for *program*)	PC for MTech (MCS) doing MTech by Dissertation and Dual(CS)

7.	**Pre-requisites** (course no./title)	COD892

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Every sem
11.	Faculty who will teach the course - All	
12.	**Will the course require any visiting faculty?**	May be

13.	**Course objective** *(about 50 words)*: To continue the work of MTP-I COD892 to completion and prepare dissertation on the problem and its solution.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The student(s) who work on a project are expected to work towards the goals and milestones set in CSD851. At the end there would be a demonstration of

	the solution and possible future work on the same problem. A dissertation outlining the entire problem, including a survey of literature and the various results obtained along with their solutions is expected to be produced by each student.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	INDEPENDENT STUDY
3.	**L-T-P structure**	0-3-0
4.	**Credits**	4
5.	**Course number**	COS799
6.	**Status** (*category* for *program*)	PE for MTech (MCS)/CS5/CSY/SIY

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	

9.	**Not allowed for** (indicate program names)	CS1, ANZ/CSZ

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	ANY
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: The course is intended to give the students a self-directed independent research and

	problem solving in a area of Computer Science.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The student will be tasked with certain reading assignments and related problem solving in a appropriate area of research in Computer Science under the overall guidance of a CSE Faculty member. The work will be evaluated through term paper.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
COURSE TOTAL *(14 times 'L')*		

16.	**Brief description of tutorial activities: Student will be assigned reading and tutorial tasks including paper reading and problem solving.**

17.	**Brief description of laboratory activities**

Module no.	**Experiment description**	**No. of hours**
COURSE TOTAL *(14 times 'P')*		

18.	**Suggested texts and reference materials**

1.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	10%
20.2	Open-ended problems	N.A.
20.3	Project-type activity	10%
20.4	Open-ended laboratory work	20%
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module on Visual Computing
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV 877
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every sem 1st sem 2nd sem Either sem

11.	**Faculty who will teach the course**	Prof Subhashis Banerjee Prof PremKalra Prof Subodh Kumar
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: This course is meant to be a module with emphasis on some research topics in the broad area of visual computing.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The course will be a seminar-based course where the instructor would present topics in a selected theme in the area of visual computing through research papers. Students will also be expected to participate in the seminar.

15.	**Lecture Outline***(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
COURSE TOTAL *(14 times 'L')*		

16.	**Brief description of tutorial activities**

N.A.

17.	**Brief description of laboratory activities** The course structure is 1-0-0, however to reinforce some concepts it may be desired that students implement and experiment through assignments/projects in the course.

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
COURSE TOTAL ***(14 times 'P')***		

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. **Research Papers**

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Matlab/OpenCV/OpenGL for doing assignments/project.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	Standard
19.4	Laboratory	Standard
19.5	Equipment	Standard
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	20%
20.3	Project-type activity	20%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Note: The course is designed to be a mix of theory as well as hands on experience. A background in Linear Algebra and Basic Probability is required.

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Machine Learning
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV878
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites**(course no./title)	---

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (Indicate program names)	---

10.	**Frequency of offering**	Either sem

11.	**Faculty who will teach the course**	K KBiswas, Subhashis Banerjee, ParagSingla, Mausam, ManikVarma
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: This course is meant to be a Special Module in Machine Learning. The objective of this course is to provide anin depth exposure to a particular sub-area of Machine Learning. A set of advanced topics in Machine Learning would be covered depending on instructor’s expertise and interests.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Contents may vary based on the instructor's expertise and interests within the broader area of Machine Learning.Example topics include (but not limiting to) Statistical Relational Learning, Markov Logic, Multiple Kernel Learning, Multi-agent Systems, Multi-Class Multi-label Learning, Deep Learning, Sum-Product Networks, Active and Semi-supervised Learning, Reinforcement Learning, Dealing with Very High-Dimensional Data, Learning with Streaming Data, Learning under Distributed Architecture

15.	**Lecture Outline***(with topics and number of lectures)* Exact Outline will depend on the instructor and the specific content of the course being offered.

16.	**Brief description of tutorial activities** - NA
17.	**Brief description of laboratory activities** - NA

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Kevin P. Murphy. **"**Machine Learning: A Probabilistic Perspective". MIT Press, 2012.
2. Christopher Bishop. "Pattern Recognition and Machine Learning", First Edition,Springer, 2006.
3. Daphne Koller and Nir Friedman. "Probabilistic Graphical Models: Principles and Techniques". MIT Press, 2009.
4. R. Duda, P. Hart and D. Stork. "Pattern Classification" Second Edition, John Wiley & Sons INC., 2001.
5. Tom Mitchell. "Machine Learning" First Edition, McGraw-Hill, 1997.
6. Relevant Research Papers.
7. Online Reading Material.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	C/C++/Java. Perl. Free Download (e.g. Weka machine learning repository, UCI datasets). Matlab.
19.2	Hardware	Personal Laptop or Computer Access
19.3	Teaching aides (videos, etc.)	N.A.
19.4	Laboratory	N.A.
19.5	Equipment	N.A.
19.6	Classroom infrastructure	Data Projector
19.7	Site visits	N.A.

20.	**Design content of the Course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	40%
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Financial Algorithms
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV879
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS
7.	**Pre-requisites** (course no./title)	MAL 250 or MAL 140 (old)

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	About 20% overlap with
8.2	Overlap with any UG/PG course of other Dept./Centre	MAL 732 & MAL 733
8.3	Supercedes any existing course	No

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course -** Nil Dr. Prasoon Tiwari	
12.	**Will the course require any visiting faculty?**	Dr. Prasoon Tiwari B65 SFS Sheikh Sarai Phase-I New Delhi 110017

13.	**Course objective** *(about 50 words)*: The module will start with model independent results in replication of derivatives, and then move on to pricing of derivatives using the binomial method followed by Black-Scholes formula derivation via stochastic calculus. Various Financial Algorithms for Statistical arbitrage, structured products and volatility derivatives will be discussed and the module will conclude with introduction to martingale methods.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The indicative contents will be from a mix of the following Model Independent Results, Vanilla/Exotic Options and Binomial Pricing, Stochastic Calculus, Black-Scholes, Statistical Arbitrage, Structured Products, VIX, Martingales, Trading Strategies

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Introduction	1
2	Model Independent Derivative Replication	1
3	Binomial Pricing of Vanilla and Exotic Derivatives	2
4	Brownian Motion	1
5	Stochastic Calculus	1
6	Black-Scholes Formula	2
7	Statistical Arbitrage	2
8	Structured Products	1
9	VIX	1
10	Martingale Methods	1
11	Trading Strategy Presentation	1
	COURSE TOTAL	14

16. Brief description of tutorial activities

Students will be asked to come up with a market scenario and a trading strategy for that market scenario.

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
	No specific lab activities for this project.	
	COURSE TOTAL *(14 times 'P')*	

18. Suggested texts and reference materials

1. Options, Futures, and Other Derivatives by John C. Hull.
2. Class handouts.
3. Option Markets by John C. Cox and M. Rubinstein.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	Computers
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	Standard software development lab will do.
19.5	Equipment	
19.6	Classroom infrastructure	Projector for PowerPoint Slides (i will bring my own laptop)
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	15% Students will define a market scenario and design a trading strategy for it. This will be presented in class.
20.2	Others (please specify)	85% Lectures

Date: **(Signature of the Head of the Department)**

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Parallel Computation
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV880
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	Permission of Instructor

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	
8.2	Overlap with any UG/PG course of other Dept./Centre	
8.3	Supersedes any existing course	

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Either Sem

11.	**Faculty who will teach the course**	Subodh Kumar, Rahul Garg, Sandeep Sen
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in the area of parallel computation.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special module that focuses on special topics and research problems of importance in this area

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
COURSE TOTAL ***(14 times 'L')***		**14**

16.	**Brief description of tutorial activities**

17.	**Brief description of laboratory activities:**

Module no.	**Experiment description**	**No. of hours**
1		
2		
COURSE TOTAL ***(14 times 'P')***		

18.	**Suggested texts and reference materials** STYLE: Author name and initials, Title, Edition, Publisher, Year.

1. Research Papers

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	.
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Harware Systems
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV 881
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	Permission of Instructor

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year
11.	Faculty who will teach the course- Anshul, Mbala, Kolin, Preeti, Smruti	
12.	**Will the course require any visiting faculty?**	Yes
13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area	
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special module that focuses on special topics and research problems of importance in this area	

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		14

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Software Systems
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV 882
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	Every other sem

11.	**Faculty who will teach the course** Huzur Saran, Sanjiva Prasad, S. Arun Kumar, Sorav Bansal, S.C. Gupta, Subodh Kumar, Kolin Paul, Smruti Ranjan Sarangi

12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in the area of computer systems, esp. Software systems. Topics may include, but are not limited to, OS, Compilers, Networking software, Virtualization, Cloud Computing,

	Distributed Computing, Parallel Computing, Heterogeneous Computing, etc.
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special topic that focuses on state of the art and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1	Lectures contents will depend on the exact topics being covered. They will typically involve discussing recent and classic research papers in the area	
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		14

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Moduleno.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

Research papers, online material. Perhaps, some advanced books, but that will depend on the topics being covered.

19. **Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	Usually Open-source Software
19.2	Hardware	One shared lab computer per student
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. **Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Theoretical Computer Science
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV 883
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	CSL 351 (Algorithm Design and Analysis) or equivalent

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	NO
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐Every sem ☐1st sem ☐2nd sem ☒Alternate year
11.	Faculty who will teach the courseAll	
12.	**Will the course require any visiting faculty?**	
13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area	
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special module that focuses on special topics and research problems of importance in this area	

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		14

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science & Engineering
2.	**Course Title** *(< 45 characters)*	SPECIAL MODULE IN ALGORITHMS
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV886
6.	**Status** (*category* for *program*)	Post-Graduate-PE/UG-OC

7.	**Pre-requisites** (course no./title)	CSL 351 (Algorithm Design and Analysis) or equivalent

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	No
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	
10.	**Frequency of offering**	☐ Every sem ☐ 1st sem ☐ 2nd sem ☒ Alternate year

11.	Faculty who will teach the courseAll	
12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area
14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special module that focuses on special topics and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	Experiment description	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Departmen

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Artificial Intelligence
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV 884
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL333or COL671

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No
8.3	Supersedes any existing course	No

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	Mausam, Saroj Kaushik, K K Biswas, Parag Singla
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: The students will learn about topics, which are currently at the forefront of Artificial Intelligence research. Each semester, the theme of the course may change depending on the instructor.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Potential topics or themes which may be covered (one topic per offering) include: advanced search, applications of AI to crowdsourcing, document summarization, game playing, reinforcement learning, etc.

15.	**Lecture Outline***(with topics and number of lectures)*: No specific list of lectures can be specified since the topic of the course changes from offering to offering.

16.	**Brief description of tutorial activities:** NA

17.	**Brief description of laboratory activities**: NA

18.	**Suggested texts and reference materials**

1. Papers published in conferences and journals will be suggested based on the topic.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	NA
19.2	Hardware	NA
19.3	Teaching aides (videos, etc.)	NA
19.4	Laboratory	NA
19.5	Equipment	NA
19.6	Classroom infrastructure	Projector
19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Computer Applications
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV885
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS
7.	**Pre-requisites** (course no./title)	

7.	**Pre-requisites** (course no./title)	Permission of Instructor

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	---

10.	**Frequency of offering**	Every sem 1st sem 2nd sem Alternate year

11.	**Faculty who will teach the course**	
12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special topic that focuses on special topics and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Module no.	Topic	No. of hours
1		
2		
COURSE TOTAL *(14 times 'L')*		14

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Moduleno.	Experiment description	No. of hours
1		
2		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

1. Research Papers

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Algorithms
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COL 886
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre-requisites** (course no./title)	COL 351 (Analysis and Design of Algorithm) or equivalent

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	No
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	☐Every sem ☐1st sem ☐2nd sem ☒Alternate year

11.	**Faculty who will teach the course**All	
12.	**Will the course require any visiting faculty?**	

13.	**Course objective** *(about 50 words)*: To provide insight into current research problems in this area

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Special module that focuses on special topics and research problems of importance in this area

15. Lecture Outline *(with topics and number of lectures)*

Modul e no.	Topic	No. of hours
1		
2		
3		
4		
5		
6		
7		
8		
9		
10		
11		
12		
COURSE TOTAL *(14 times 'L')*		14

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Modul eno.	Experiment description	No. of hours
1		
2		
3		
COURSE TOTAL *(14 times 'P')*		

18. Suggested texts and reference materials

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.3	Project-type activity	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in High Speed Networks
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV887
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL334, COL672

8.	**Status vis-à-vis other courses** *(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	N/A
8.2	Overlap with any UG/PG course of other Dept./Centre	N/A

9.	**Not allowed for** (indicate program names)	N/A

10.	**Frequency of offering**	Alternate semester

11.	**Faculty who will teach the course**	1. V. Ribeiro, 2. Huzur Saran, 3. A. Seth
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)*: Computer Networks is a rapidly evolving field. The course will introduce students to advanced topics in Networks and open up current research problems in these areas.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The course will be delivered through a mix of lectures and paper reading seminars on advanced topics in Computer Networks. Students will be introduced to topics such as software defined networking, content distribution, advanced TCP methodologies, delay tolerant networking, data center networking, home networking, green networking, clean state architecture for the Internet, Internet of things, etc.

15.	**Lecture Outline** *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	Crash course on computer networks	2
4	Modules for the chosen advanced area	12
COURSE TOTAL ***(14 times 'L')***		**14**

16.	**Brief description of tutorial activities:** NA
17.	**Brief description of laboratory activities**: NA
18.	**Suggested texts and reference materials**

1. Papers from conferences such as SIGCOMM, INFOCOM, HOTNETS, IMC, PAM, MOBICOM, MOBISYS, NSDI, CCR, CONEXT.
2. Top-down approach to computer networks, by Kurose and Ross
3. Computer networks: A systems approach, by Peterson and Davie
4. Engineering approach to computer networks, by S. Keshav
5. Patterns in network architecture, by John Day

19.	**Resources required for the course** *(itemized & student access requirements, if any)*
20.	**Design content of the course** *(Percent of student time with examples, if possible)*

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Database Systems
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV 888
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS

7.	**Pre Requisites** (course no./title)	COL362

8.	**Status vis-à-vis other courses***(give course number/title)*	
8.1	Overlap with any UG/PG course of the Dept./Centre	No
8.2	Overlap with any UG/PG course of other Dept./Centre	No
8.3	Supersedes any existing course	No

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Either semester

11.	**Faculty who will teach the course**	S.K. Gupta, Maya Ramanath
12.	**Will the course require any visiting faculty?**	**No**

13.	**Course objective** *(about 50 words)***:** The students will learn about topics, which are currently at the forefront of Database and Data Management research. Each semester, the theme of the course may change depending on the instructor.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: Potential topics or themes which may be covered (one topic per offering) include: data mining, big data management, information retrieval and database systems, semantic web data management, etc.

15.	**Lecture Outline***(with topics and number of lectures)*: No specific list of lectures can be specified since the topic of the course changes from offering to offering.

Module no.	Topic	No. of hours
		14

16.	**Brief description of tutorial activities: NA**

17.	**Brief description of laboratory activities: NA**

Module no.	Experiment description	No. of hours

18.	**Suggested texts and reference materials**

1. Papers published in conferences and journals will be suggested based on the topic.

19.	**Resources required for the course** *(itemized & student access requirements, if any)*

19.1	Software	NA
19.2	Hardware	NA
19.3	Teaching aides (videos, etc.)	NA
19.4	Laboratory	NA
19.5	Equipment	NA
19.6	Classroom infrastructure	Projector

19.7	Site visits	N.A.

20.	**Design content of the course** *(Percent of student time with examples, if possible)*

20.1	Design-type problems	
20.2	Open-ended problems	
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)

COURSE TEMPLATE

1.	**Department/Centre proposing the course**	Computer Science and Engineering
2.	**Course Title** *(< 45 characters)*	Special Module in Concurrency
3.	**L-T-P structure**	1-0-0
4.	**Credits**	1
5.	**Course number**	COV889
6.	**Status** (*category* for *program*)	PE for CS5, CSZ, CSY, SIY, ANZ, MCS
7.	**Pre-requisites** (course no./title)	MAL 250 or MAL 140 Iold)

7.	**Pre-requisites** (course no./title)	---

8.	**Overlap of contents with any** *(give course number/title)*	
8.1	existing UG course(s) of the Department/Centre	
8.2	proposed UG course(s) of the Department/Centre	
8.3	approved PG course(s) of the Department/Centre	
8.4	UG/PG course(s) from other Departments/Centers	
8.5	Equivalent course(s) from existing UG course(s)	

9.	**Not allowed for** (indicate program names)	

10.	**Frequency of offering**	Every sem 1st sem 2nd sem Alternate year

11.	**Faculty who will teach the course** Sanjiva Prasad, S. Arun Kumar	
12.	**Will the course require any visiting faculty?**	No

13.	**Course objective** *(about 50 words)*: Make students familiar with the key notions of concurrency theory, models of concurrent systems, Specificational methods, Tools and techniques.

14.	**Course contents** *(about 100 words) (Include laboratory/design activities)*: The course will focus on research issues in concurrent, distributed and mobile computations. Some of the following topics will be covered: Models of Concurrent, Distributed and Mobile computation. Process calculi, Event Structures, Petri Nets an labeled transition systems. Implementations of concurrent and mobile, distributed

	programming languages. Logics and specification models for concurrent and mobile systems. Verification techniques and algorithms for model checking. Type systems for concurrent/mobile programming languages. Applications of the above models and techniques.

15. Lecture Outline *(with topics and number of lectures)*

Module no.	**Topic**	**No. of hours**
1	The exact lecture contents will depend on the topics being covered in that year.	
2		
	COURSE TOTAL *(14 times 'L')*	14

16. Brief description of tutorial activities

17. Brief description of laboratory activities

Module no.	**Experiment description**	**No. of hours**
1		
	COURSE TOTAL *(14 times 'P')*	

18. Suggested texts and reference materials

STYLE: Sonntag, R. E., Borgnakke, C., and Van Wylen, G. J., *Fundamentals of Thermodynamics*, 5th Ed., John Wiley, 2000.

Various recent books on concurrent systems, papers and materials available on the web.

19. Resources required for the course *(itemized & student access requirements, if any)*

19.1	Software	
19.2	Hardware	
19.3	Teaching aides (videos, etc.)	
19.4	Laboratory	
19.5	Equipment	
19.6	Classroom infrastructure	
19.7	Site visits	

20. Design content of the course *(Percent of student time with examples, if possible)*

20.1	Design-type problems	15%
20.2	Open-ended problems	15%
20.3	Project-type activity	15%
20.4	Open-ended laboratory work	
20.5	Others (please specify)	

Date: (Signature of the Head of the Department)